**ERC Starting Grant 2024**
**Research Proposal [Part B2]**

# Designer Condensates for Regulation of Catalytic Processes

## Section a. State of the art and objectives

### 1. Synopsis

Living cells have evolved to provide subcellular microenvironments, or compartments that accommodate and control specific (bio)catalytic processes. These compartments include membrane-bound and membraneless organelles, or biomolecular condensates[1]. New breakthroughs in studying and characterizing cellular biomolecular condensates have facilitated researchers to design synthetic condensates that are formed by liquid-liquid phase separation (LLPS) of engineered proteins[2–5], polypeptides[6–8], peptides[9–16], RNA[12,17], DNA[18,19] or biopolymers[20–22]. Yet, a precise regulation of biomolecular condensates properties and their utilization for nanobiotechnological applications remains a major challenge. In this research project, I propose to utilize my expertise in peptide design to bridge this gap by developing libraries of synthetic biomolecular condensates with systematically tunable chemical composition using minimalistic LLPS-promoting peptide building blocks. I expect to gain insights into how the chemical composition of condensates affect their physical and material properties (Aim 1). I will harness this knowledge for regulation of organic reactions in engineered condensate microenvironments (Aim 2) and to develop condensates with emergent collective catalytic capacity (Aim 3) for synthesis of chemotherapeutic anti-cancer drugs (Aim 2-3). Altogether, the proposed research will elucidate how the chemical composition, physical and material properties of designed condensates affect reaction rate and conversion, and condensate reactivity.

The designed condensates will go beyond the state of the art of synthetic nano- and microreactors and provide the next level of reaction regulation for sustainable drug synthesis in an organic solvent-free, aqueous environment. Thus, this approach will open new opportunities in biotechnology for the pharmaceutical, cosmetics, and food-tech industries. The findings will contribute to the establishment of new design principles for the fabrication of synthetic condensates from peptide building blocks with ordered/disordered motifs, and for controlling and balancing these motifs to control reactivity. I expect that this research will facilitate the LLPS, peptide self-assembly, and nanobiotechnology communities for the development of condensates for new nanotechnologies.

### 2. Background and state of the art

Current synthetic approaches for preparing nano- and micro-reactors for regulation of catalytic processes fall into two broad categories: (i) systems relying on liposomes[23–25] polymersomes[26,27], water/oil emulsions[28], or reverse micelles[29]; and (ii) nano/micro-cages formed by self-assembly of DNA[30], proteins, or designed peptides[31]. Each type provides several benefits, alongside shortcomings. In particular, compartments of the first type have been extensively characterized and show promise for various biomedical applications[23,26–28]. Yet, the building blocks of these compartments have limited biodegradability and limited versatility in terms of their chemical composition. In contrast, compartments formed by cages of DNA[30], proteins, or peptides[31] are versatile, biocompatible and biodegradable. However, as closed compartments, they are limited in terms of permeability to reactants and typically have low water content which might reduce the efficiency of specific reactions. Crucially, **there is an unmet need for tuning the properties of compartments from both categories to regulate reactant encapsulation, reactivity, and conversion.**

In living systems, this need is addressed by cellular organelles, as most (bio)catalytic processes are compartmentalized in the living cell by either membrane-bound[32] or membraneless organelles[1], where they are strictly regulated both spatially and temporally. Membraneless organelles, or biomolecular condensates, are intracellular condensed phases that are formed by liquid-liquid phase separation (LLPS) of intrinsically disordered proteins (IDPs), either alone or in complexation with nucleic acids[1]. Accumulating evidence show that various IDPs and nucleic acids commonly undergo LLPS through multiple weak non-covalent interactions including H-bonding[33,34], dipole–dipole, electrostatic interactions, π-π stacking, and cation-π interaction[35–37].

Developments in the emerging field of LLPS, coupled with new breakthroughs in studying and characterizing cellular condensates, have provided the research community with the capability to design synthetic biomolecular condensates for a variety of applications, including intracellular delivery[38], storage of biomolecules[39], protein transcription-translation[4041,42] and regulation of biocatalytic processes[6,43–46]. Specifically, synthetic biomolecular condensates can be designed from a diverse set of molecular building blocks[6] including single- or multicomponent engineered proteins[2–5], polypeptides[6–8], peptides[9–16], RNA[12,17], DNA[18,19] or biopolymers[20–22]. **Yet, expanding the**

**nanobiotechnological potential of designed condensates and gaining a higher level of control over their physicochemical and material properties remains a major challenge** due to the limited available characterization techniques at the molecular-level[47] and incomplete understanding of the underlying interactions that govern their formation. There is an acute need to further explore and expand the function space of designed biomolecular condensates for applications beyond those of biological condensates. Thus, **I propose to develop condensates that encompass new nanotechnological functions beyond biological mimicry for regulation of organic reactions in aqueous environment.**

For this, I propose to use LLPS-promoting peptides as building blocks of condensates since compared with protein production, their synthesis is simple and facile, can be readily scaled, and does not require complex expression/purification steps. From a technological perspective, peptide assemblies have the potential to interface seamlessly with living systems[48], and have a greater potential for biotechnological applications than synthetic polymers, since their alpha amino acid degradation products are designated "generally recognized as safe" (GRAS). Importanty, the side-chain groups of the amino acids in peptides, provide a diverse set of simple chemical functionalities that collectively constitute a rich and versatile chemical space[49,50]. This chemical versatility can be utilized for a variety of application. For instance, I previously showed that peptides with tunable level of supramolecular order/disorder[51] can be utilized for applications in photoprotection **(Science 2017, Angew. Chem. 2021, ACS Appl. Mater., 2022)**[51–53] and conductive materials **(Adv. Mater. 2020**)[54], and materials whose properties change in response to chemical **(Angew. Chem., 2021, Adv. Funct. Mater. 2022)**[52,55,56] or physical stimuli **(ACS Appl. Mater. Interfaces 2022, Adv. Mater 2022)**[15,53]. Lastly, due to their short length and lack of secondary structure compared with proteins, utilizing peptides as building blocks facilitates the establishment of sequence-structure and structure-function relations[50], which is a key element towards understanding how to regulate the properties of condensates. Realizing the potential of designed peptide condensates, several groups, including us **(Adv. Mater., 2022, Nat. Commun., 2023, Chem. Comm., 2023, PNAS, 2023)**[14,15,56,57], have developed peptide-based condensates for a variety of applications[9–12,58–61]. Despite this progress, controlling and regulating the properties e.g., size, dispersity, architecture, stability, and viscosity of peptide condensates is still a challenge. Moreover, tailoring the properties of condensates to regulate the reactivity in the condensed phase has not been realized to date.

Over the past few years, I have gained extensive experience in developing peptide-based bioinspired materials. I have demonstrated that peptide supramolecular order/disorder can be designed, tuned, and leveraged to expand the repertoire of material properties[51,52,54,62,63]. Moving from hierarchical and thermodynamically stable biomolecular assemblies to dynamic and metastable ones, my group has been developing tools to design and characterize peptide condensates[14,15,56,57] (Figs. 2-4). One of our main focuses is gaining mechanistic understanding of the chemical and molecular basis of LLPS and utilizing this knowledge to regulate the properties of condensates for new functions. Our current findings indicate that the peptide sequence controls the LLPS behavior and condensate properties, including diffusion and encapsulation efficiency. The proposed project will build on these findings, as well as on my expertise in peptide design tools and characterization of biomolecular assemblies at the material properties level and the molecular level.

In this research project, I propose to **go beyond the state of the art** and utilize our advanced peptide design capabilities to develop **libraries of synthetic biomolecular condensates** that are formed by LLPS of minimalistic peptide building blocks for regulation of organic reactions. By using systematic sequence design approach using libraries of condensates with tunable chemical composition, I expect to

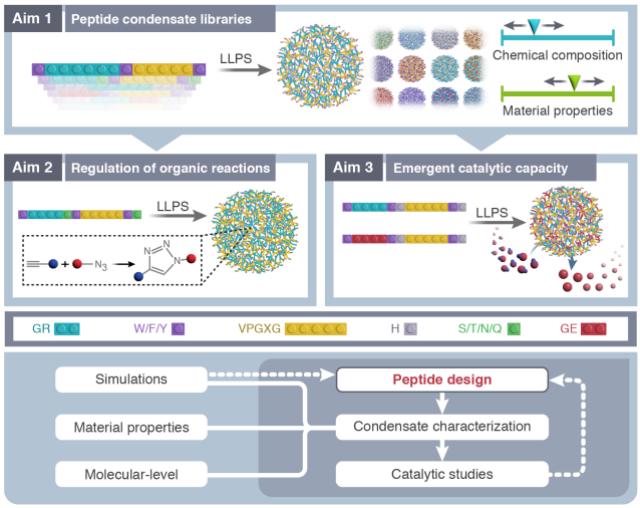


**Fig. 1**. Schematics of the specific aims and workflow of *CoRe,* which focus on the design of peptide condensates with tunable chemical and physical properties (**Aim 1**) for regulation of chemical reactions (**Aim 2**) and emergent reactivity (**Aim 3**). Lego blocks represent amino acids motifs that will be incorporated to the peptide sequences. **Color code**: LLPS promoting motif GR (turquois), LLPS anionic motif GE (red), aromatic amino acids (purple), aliphatic elastin-like polypeptide motif (yellow), histidine (grey), polar amino acids (green). **Project workflow** (lower panel): the research is expected to be highly iterative: the findings from the characterization of condensates, simulations, catalytic studies, and molecular level studies will feed back (dashed arrows) into the peptide design process for further cycles of design optimization.

**gain insights on how to precisely control and tune the physicochemical properties of the condensed phases**. I then will harness these insights to regulate reactant partitioning, reaction rate and conversion in organic solvent-free environment for synthesis of chemotherapeutic anticancer drugs. Moreover, I aim to develop condensates with **emergent collective catalytic capacity** that mimic the catalytic activity of hydrolyses, to catalyze the synthesis of small molecule anticancer drugs, by **regulating the reactivity** of the peptide in the condensed phase. This approach will offer **mechanistic understanding** of how the physical and materials properties of condensates affect reactivity in the condensed phase.

## 3. Project Objectives:

In this work, I will go beyond the current state of the art[8,9,12,16,58,64–70] **and develop libraries of designed synthetic peptide condensates with tunable chemical and physical properties (Aim 1). I will utilize these libraries for regulation of catalysis, either by providing a controlled microenvironments for organic reactions (Aim 2) or by emergent catalytic capacity and controlled reactivity (Aim 3)**. I will exploit a systematic sequence design approach and develop libraries of condensates with varying chemical composition.

The underlying hypotheses of *CoRe* are: (i) Synthetic condensates with tailored physicochemical properties can be engineered from LLPS-promoting peptide building blocks; (ii) the capacity to control, regulate, and tune reactivity is determined by the chemical and materials properties of condensates, which are encoded in the peptide sequence.

To achieve the goal of *CoRe*, I will pursue the following specific aims:

Aim 1: **Develop synthetic customizable biomolecular condensates with controlled architecture, chemical and material properties and elucidate sequence-structure relations underlying condensate formation.** We will use peptides with LLPS-promoting motifs to engineer compartments that regulate the partitioning of hydrophobic substrates and host biocatalytic reactions.

Aim 2**: Regulation of organic reactions by controlled microenvironments using designer condensates.** We will employ the libraires of synthetic peptide condensates developed in Aim 1 to regulate organic reaction in organic-solvent free conditions **by regulating their chemical composition and material properties.**

Aim 3: **Develop peptide condensates with emergent catalytic capacity and controlled reactivity**. We will employ the libraries of synthetic peptide condensates developed in Aim 1 and investigate emergent catalytic capacities of designed condensates. We will elucidate how the **chemical and physical properties of the condensates affect their reactivity.**

## Significance and impact:

*CoRe* is an ambitious project that proposes a new approach to the far-reaching and cross-disciplinary puzzle of reaction compartmentalization and mimicry of cellular organelles. This is project combines chemistry, materials science, and biotechnology, to provide fundamentally new strategies for controlling and regulating the properties of biomolecular condensates, which are emerging liquid assemblies, and utilizing them as reaction vessels, for drug synthesis in organic solvent-free and cell-free environments. Accordingly, as described further below, the findings in this project are expected to promote scientific progress in the fields of biotechnology, biomaterials, and supramolecular chemistry; and to have a significant impact on relevant industries.

**Biotechnological impact:** Our peptide LLPS-based approach for condensate production opens new opportunities for sustainable drug synthesis in organic solvent-free aqueous environment, in accordance with the guidelines of green chemistry. The libraries of condensates produced in this project are expected to provide tools for regulation of organic reactions and enzyme-free emergent catalytic capacity for drug synthesis. Since the peptide building blocks degrade into alpha-amino acids, unlike cell-based microreactors and drug synthesis systems, the condensates can be readily integrated in formulations with no need for purification steps. This research will lay the groundwork for versatile and customizable condensate technologies that offer tremendous commercial opportunities for the pharmaceutical, cosmetics, and food tech industries. For example, the designed condensates will provide bio-friendly, organic solvent-free microenvironments for chemical synthesis of chemotherapeutic drugs (Aim 2 and 3) as well as enzyme-free hydrolase activity for food tech and cleantech applications (Aim 3).

**Impact on the scientific community**: The findings of this work, will yield an exhaustive array of characterization analyses, ranging from material-level to molecular-level studies and will significantly enhance our understanding of the underlying interactions governing peptide LLPS. Crucially, our findings will contribute to the establishment of **design principles** for the fabrication of peptide building blocks with ordered/disordered motifs and suggest methods to control and balance such motifs to achieve specific properties, reactions, and capabilities—including

controlled compartmentalization, reactivity, and drug production, which are of tremendous importance for healthcare and biotechnological applications. As we plan to explore the mechanism of catalysis in/by the condensates, the findings from the aims including the intermediate goals will provide important insights on how the building block sequence affect catalytic performance and emergent properties of condensates as catalysts. These insights will facilitate the development of condensates for nanotechnological applications, namely for green chemistry, and can be expanded for development of renewable energy technologies.

## Section b. Methodology

**Aim 1:** Develop synthetic peptide condensates with controlled architecture, chemical, and material properties and elucidate sequence-structure relations underlying condensate formation.

1.1 Design and synthesis of peptide condensate libraries: We will develop 5 libraries of peptide condensates with systematically tunable chemical composition: **Library 1** will include LLPS peptides with varying aromatic amino acid content; **Libraries 2** will include LLPS peptides with varying hydrophobicity; **Library 3** will include LLPS peptides with varying polarity; **Libraries 4** will include LLPS peptides with varying length, which, in complexation with anionic biopolymers, form condensates with tunable architecture; and **Library 5** will include LLPS peptides with varying number and position of histidine (H) as a catalytic moiety.

Our design approach is a combination of bioinspiration from low complexity protein domains (LCDs) of IDPs and rational design based on the chemical properties of the amino acids side chains. We recently showed that minimalistic variants of LCDs phase separate into liquid condensates under specific conditions (i.e., concentration, pH and ionic strength)[14]. **We will build on our recently developed library of LLPS-promoting peptides that form condensates with tunable dynamics and encapsulation efficiency (Nat. Commun., 2023)**[14]. These peptides contain alternating glycine-arginine (GR) dyads, which are abundant in LCDs of IDPs and provide charge (R) and backbone flexibility (G), as well as the aromatic amino acids tryptophan (W) and tyrosine (Y), and the

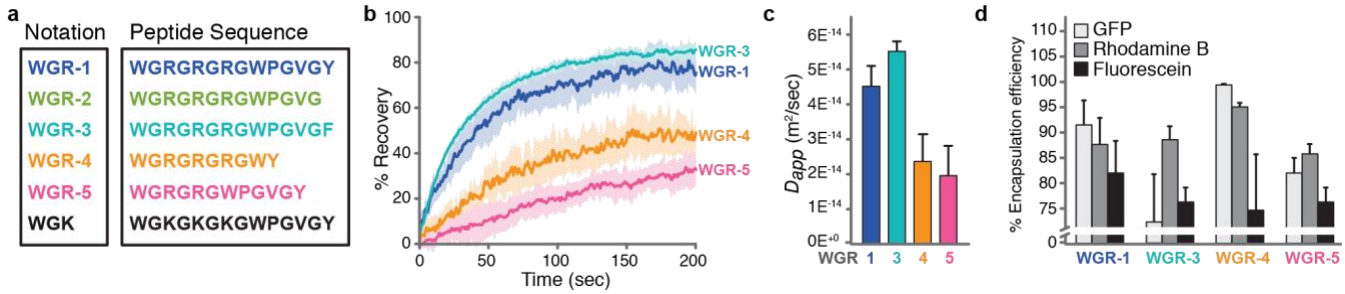


**Fig. 2. Peptide sequence controls LLPS propensity and condensate material properties (Nat. Commun., 2023)**[14]. **a**. Library of LLPS-promoting peptide sequences. **b-c**. Recovery plots (**b**) and apparent diffusion coefficient $D_{app}$ (**c**) obtained from FRAP analysis using FITC-labeled peptides. Values represent average n=7, error bars are indicated **d**. Encapsulation efficiency of fluorescent payloads in peptide condensates, analyzed by absorbance measurements. Values represent an average of n=3, error bars are indicated.

repeating elastin-like polypeptide (ELP) motif VPGXG[71,72], with V replaced by W. Our recent results indicate that the peptide sequence controls LLPS propensity[14] as well as material properties of the resulting condensates including peptide mobility and diffusion (Fig. 2b-c), and the encapsulation efficiency of fluorescent payloads (Fig. 2d). Our results also suggest a critical role for π-interactions between the side chains of R and that of the aromatic amino acid side chains in LLPS. While both K and R can participate in cation-π interactions, only R can form π-π interactions due to the $sp^2$ nitrogen atoms. Based on these results, we will employ two approaches to promote LLPS: (i) simple coacervation triggered by charge screening via pH and ionic strength tuning; (ii) complex coacervation of the peptides with either polyuridylic acid (poly-U), which is not prone to form a secondary structure[12], or heparin as a model system of anionic biopolymers.

*The peptide sequences will include the following motifs*:

1) *Alternating GR motif.* This repeating motifs is found in the LCD of membraneless organelle-forming IDPs and is known to promote coacervation in vitro[73], with the G providing the backbone flexibility important for LLPS, and the R providing charge and promoting electrostatic and π-π interactions.

2) *Aromatic amino acids*. Building on previous reports that the guanidinium side chain of R participates in cation-π[74] or π-π interactions with aromatic amino acids, we will incorporate aromatic amino acids that can interact with the side chain of R.

3) *Aliphatic amino acids*. To promote both coacervation and interactions with hydrophobic substrates, we will incorporate aliphatic amino acids, either alone or as part of the hydrophobic elastin-like polyprotein (ELP) repeating domain VPGXG, where we systematically study the effect of the amino acids at the X position on LLPS and substrate partitioning. The ELP motif promotes phase separation, and hydrophobic amino acids induce hydrophobic microenvironment in the condensate which might promote partitioning of hydrophobic reactants.
4) *Polar amino acids*. These will be integrated in the peptide sequences to promote H-bonding with the solvent to facilitate hydration of the condensates, or peptide-peptide H-bonding.
*For comparative analysis between peptides from different libraries, the net charge of all peptides will be kept (+3).

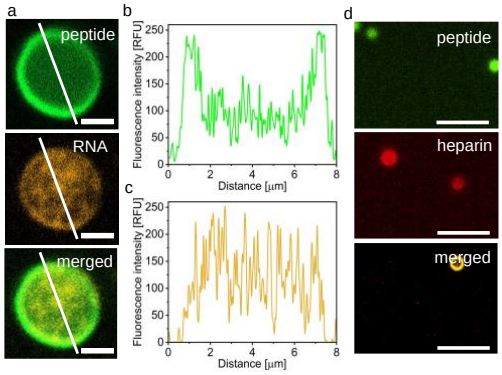


**Fig. 3. Regulation of condensate architecture in peptide-RNA (PNAS, 2023, *in press*)[56] or peptide-biopolymer systems (Unpublished preliminary results). a.** Confocal microscopy images of FITC-labelled peptide (green)-polyU (Cy3, orange) condensates ($\lambda_{ex}$=488 nm and $\lambda_{ex}$=562 nm). Scale bars=2.5 μm. **b-c**. Fluorescence intensity line profiles of (**b**) FITC-labeled peptide and (**c**) Cy3-oligoA. **d**. Fluorescence microscopy images of FITC-peptide/Atto647 heparin condensates. Scale bars=5 μm.

**We will develop 5 libraries where in each library we systematically tune a different chemical property (Table 1).** The libraries will be later utilized for regulation of catalysis (Aim 2 and Aim 3) by providing tunable microenvironments:
i. In **Library 1** the aromatic amino acid composition of the condensates is tuned. We hypothesized that the aromatic amino acids in the condensate promote interactions with aromatic reactants and thus facilitate their partitioning.
ii. In **Library 2** the hydrophobicity of the condensates is tuned by systematically varying the composition of hydrophobic amino acids: V, L, I, M, and their position in the peptide. We expect that larger branched chain amino acids (BCAA) will facilitate peptide-peptide interactions more than smaller BCAA e.g., V. Using these libraries, we aim to find the optimized composition of aliphatic amino acids which promotes partitioning of hydrophobic reactants and their chemical transformation while maintaining LLPS. Our preliminary results (Fig. 6; Aim 2) show that the presence of 2 aliphatic amino acids significantly attenuates the kinetics of click reaction and decrease conversion, thus we will only use a single aliphatic amino acid.
iii. In **Library 3** the polarity of the condensates is tuned by systematically varying the number (1 vs. 2) and composition of the polar amino acids: S, T, Q, N. Based on our preliminary results (Fig. 6) we hypothesized that the polarity/hydrophobicity has an important role in regulation of chemical reactions. We will test this hypothesis in Aim 2 by using Libraries 2-3.
iv. In **Library 4** we aim to tune the architecture of the condensates. Here, we rely on our recent preliminary results (Fig. 3) which show that the interaction of short peptides containing high density of aromatic amino acids (>2) phase separate into condensates with a core-shell morphology (PNAS 2023, *in press*)[56] through complexation with poly-U (Fig. 3a-c) or with heparin as an anionic biopolymer (Fig. 3d; unpublished), where the peptide is mostly concentrated in the condensate shell and the RNA is distributed throughout the condensate (Fig. 3a-c). Building on these early results, we will tune the architecture of the condensates by complexation of the peptides in Library 4 with poly-U or with heparin (*N*=70). Specifically, we will examine how the peptide length and composition of the basic amino acid R vs. K affect the complexation with RNA/heparin and the resulting architecture (more details in Aim 1.3). We hypothesize that decreasing the peptide length (while keeping a constant number of aromatics) increase strength of peptide-peptide interactions vs. peptide-RNA or peptide-heparin interactions, and this, in turn, will facilitate formation core-shell architectures. We will examine if we can tune the diameter of the condensate shell vs. core by varying peptide length and composition.
v. In **Library 5** we aim to tune the catalytic capacity of the condensates by varying the position and number of histidine (H) in the peptide sequence. We rely on our preliminary results (Fig. 8; Aim 3) which show that condensates formed by H-containing LLPS peptides enhance ester hydrolysis mediated through metal binding. Thus, we will examine how the composition of the peptides and number and position of H affect the emergent catalytic capacity of the condensates.

Notably: the fundings from the characterization (Aims 1.1-1.5) of each library will feedback iterative design and optimization of additional building blocks, beyond those detailed in Table 1.

| **Library 1** | **Library 2** | **Library 3** | **Library 4** | **Library 5** |
|---|---|---|---|---|
| WGRGRGRGWPGVGY | WGRGRGRGWPGVGY | WGRGRGRGWPGSGY | WGRGRGRGWPGVGY | WGRGRGRGWPGVGH |
| WGRGRGRGWPGVGF | WGRGRGRGWPGLGY | WGRGRGRGWPGTGY | WGKGKGKGKGWPGVGY | WGRGRGRGWPGHGY |
| WGRGRGRGWPGVGA | WGRGRGRGWPGIGY | WGRGRGRGWPGNGY | WGRGRGRGWPGVY | WHRGRGRGWPGVGY |
| WGRGRGRGAPGVGY | WGRGRGRGWPGMGY | WGRGRGRGWPGQGY | WGRGRGRGWPGY | WGRGRGRGWPGVHH |
| AGRGRGRGWPGVGY | WVRGRGRGWPGAGY | WSRGRGRGWPGSGY | WGRGRGRGWGY | WGRGRGRGWPGHGH |
| WGRGRGRGAPGVGA | WLRGRGRGWPGAGY | WTRGRGRGWPGTGY | WGRGRGRGWY | WHRGRGRGWPGVGH |
| AGRGRGRGAPGVGY | WIRGRGRGWPGAGY | WNRGRGRGWPGNGY | WGRGRGRWY | WHRGRGRGWPGHGH |
| AGYGRGYGAPGVGY | WMRGRGRGWPGAGY | WQRGRGRGWPGQGY | WGRGRGRW | WHRGRGRHWPGHGH |

**Table 1. Suggested libraries of condensate building blocks.** The table shows a subset of the peptide sequences that will be utilized to construct the libraries of condensates. All the analyses in Aim 1.2-1.5 **will feedback iterative cycles of design optimization and synthesis of additional building blocks.**

Peptides will be synthesized by solid phase peptide synthesis using a peptide synthesizer (available to the Lampel lab) and purified by analytical and preparative high-performance liquid chromatography (HPLC) systems at the Lampel lab. The mass will be confirmed using liquid chromatography–mass spectrometry (LC-MS).

1.2. Characterization of LLPS propensity of designed peptides: Here we will examine how the chemical composition of the peptides in each library i.e., aromatic amino acid content, hydrophobic/polar amino acid content, peptide length, R vs. K, and number of H affect LLPS propensity. For this, we will generate phase diagrams for each peptide as a function of peptide concentration, RNA/heparin concentration (in combination with peptide concentration when complex coacervation is applied), pH, or ionic strength by monitoring the optical density of the samples. For this, we will use a spectrophotometer that can measure 384-well plates, to conduct high throughput analysis of various LLPS conditions as mentioned above. We will also perform microscopy analysis (using bright field microscopy or transmitted light confocal microscopy) to confirm that the optical density is a result of condensate formation rather than aggregates or other ordered structures which scatter light.

1.3. Study the morphology and architecture of designed condensates: We will characterize the morphology and architecture of condensates from each library. For Library 4 (peptide-RNA and peptide-heparin), we will examine how the peptide length and composition affect the architecture of the condensates, where we aim to tune the diameter of the core and shell of condensates by using different peptide-RNA/heparin stoichiometry. We will use various complementary microscopy and spectroscopy characterization techniques for these analyses:

1. *Dynamic light scattering (DLS).* The size distribution of the condensates will be analyzed by DLS, which enables light scattering measurements of spherical assemblies with a diameter of up to ~10 μm (available at TAU's Nanocenter).

2. *Optical microscopy imaging*. We will analyze the morphology and size of condensates by bright field optical microscopy or transmitted light confocal microscopy and monitor condensate dynamics and coalescence over time.

3. *TEM imaging.* We will analyze the morphology and size of the condensates in the nanoscale. While analyzing condensates in the dried-state may not truly reflect their structure in solution, dehydration of condensates and fixation by negative staining prevents coalescence and thus provides information about the nanometer-sized condensates that otherwise typically co-exist and coalesce with larger micron-sized examples. TEM analysis can be especially useful for condensates with a core-shell architecture[56], as an alternative tool to analyze the electron density of the shell/core without using labeling dyes as the latter might promote aggregation or intermolecular interactions.

4. *Confocal laser scanning microscopy imaging*. We will label the peptides with a fluorescent dye (e.g., FITC labeling through the terminal amine) and label the heparin using e.g., Atto647, and use Cy5-labled oligo-A that can hybridize with poly-U to label the RNA polymer. We will perform real-time imaging on the condensates using a high-resolution confocal microscopy (Zeiss LSM 880). To analyze the core-shell morphology of condensates (Library 4), we will use z-stacking with channels compatible with the labeling dyes to analyze the localization of each building block (peptide vs. polymer) in the condensate.

1.4. Study the material properties of designed condensates: We will use the following techniques to analyze the material properties of condensates:

1. *Fluorescence recovery after photobleaching (FRAP).* We will use FRAP studies to measure the diffusion kinetics of condensates formed by labeled peptides. We have previously successfully analyzed the mobility and diffusion of peptide condensates by using FITC-labeled peptides (Fig. 2b-c, Fig. 4a-b), and Fig. 2c presents the apparent diffusion coefficients ($D_{app}$) of FITC-labeled WGR-1, WGR-3, WGR-4, and WGR-5, calculated from FRAP recovery plots, revealing sequence-dependent diffusion. Fig. 4b presents $t_{50}$ values of peptide-RNA condensates obtained from FRAP recovery plots.
2. *Microrheology analysis.* To get insights into the viscosity and diffusion of individual condensates, we will use confocal microscopy to track the mobility of fluorescent beads in condensates over time to obtain the mean square displacement (MSD) and the *D* of the beads in the condensates. The viscosity of the condensates will be calculated using the Stokes-Einstein equation. We are currently developing such microrheology protocols to analyze peptide condensate viscosity.

1.5. Molecular-level studies of designed condensates: To get sequence-structure insights how the peptide chemical composition affect LLPS and condensate properties, we will integrate both experimental and computational analyses using the following techniques:
1. *Fourier-transform infrared spectroscopy (FTIR) and circular dichroism (CD).* To examine the sequence-structure relationships of the various peptide building blocks and the underlying network of intermolecular interactions, we will analyze the assemblies in solution by FTIR and CD to learn about the peptide secondary structure and the specific vibrations associated with backbone side chain H-bonding during peptide-peptide or peptide-RNA interactions. CD analysis might be especially useful to analyze peptide-RNA interactions or effect of RNA complexation on peptide secondary structure by titrations of RNA to peptide solution. Due to scattering and high optical density, the CD analysis will be performed at sub-LLPS conditions.
2. *Wide angle x-ray scattering (WAXS)*. We will analyze freeze-dried compartments to gain insights about the network of intermolecular interactions underlying the formation of droplets. We expect that the typical condensates that are formed by LLPS (Libraries 1-3, 5) will be completely disordered and amorphous. Yet, we hypothesize that part of the core-shell condensates (Libraries 4) should have a semi-crystalline core or shell due to strong peptide-peptide interactions.
3. *Raman spectroscopy*. To gain molecular-level insights into the network of intermolecular interactions which underlie condensate formation, we will use Raman spectroscopy, by taking both average spectra of condensates in solution and that of individual condensates using Raman microscopy. We are currently developing such protocols in collaboration with Prof. Duncan Graham and Prof. Karen Faulds (University of Strathclyde) to map individual condensates and obtain spectra from selected locations in condensates[14].
4. *Nuclear magnetic resonance (NMR) analysis*. We will conduct *1D and 2D NMR imaging in collaboration with Prof. Markus Zweckstetter (German Center for Neurodegenerative Diseases, DZNE)*. We will record 2D TOCSY/NOESY experiments to sequence specifically assign non-degenerate parts of the designer peptides in non-LLPS conditions at low peptide concentrations, followed by measurement of 2D TOCSY/NOESY experiments at increasing peptide concentrations to probe intermolecular peptide interactions. To decrease the strength of potential signal broadening, 1D $^{13}C$ experiments will also be measured and analyzed.
5. *Computational studies in collaboration with Prof. Hao Dong (Nanjing University, NJU):* to get molecular level information of peptide-peptide or peptide-nucleic acid/heparin interactions for selected peptides from the different libraries, we will use both coarse-grained molecular dynamics (MD) simulations and atomistic simulation.

Preliminary results: **(i)** The design approach in this proposal builds on our recently published designer LLPS promoting peptides (**Nature**

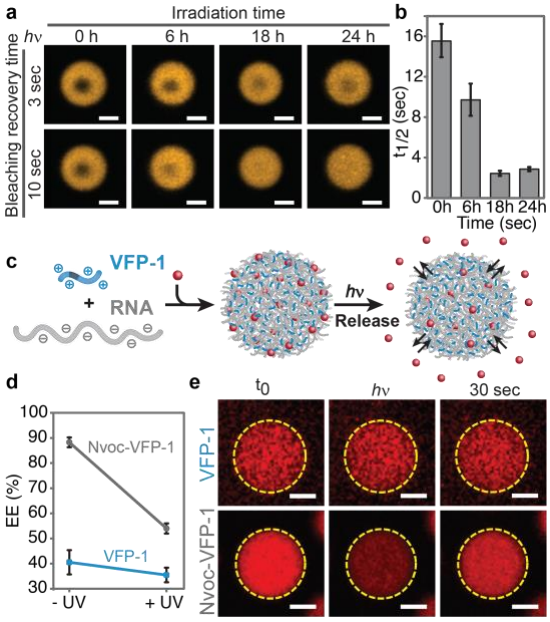


**Fig. 4. Tuning condensate material properties by light (Advanced Materials, 2022)[52].** FRAP analysis obtained by confocal microscopy of compartments formed by LLPS of model peptide VFP containing Nvoc-K (Nvoc-VFP-1) and RNA showing recovery over time (**a**) and $t_{1/2}$ values (**b**) as a function of irradiation. Values represent average of n=10, standard deviation are presented. **c**. Schematics of the designed system: irradiation of compartments that are formed by LLPS of Nvoc-VFP-1 and RNA triggers release of payload from the compartments. **d-e**. The encapsulation efficiency (**d**) and release of rhodamine B from individual compartments (**e**) is regulated by UV irradiation. Scale bar=5 μm.

**Communications**, 2023)[14], as detailed above in Aim 1.1 and Fig. 2. These are purely peptidic condensates that are formed by simple coacervation of 14-mer peptides. The results from this study show that the diffusion of the peptide in the condensate is sequence-dependent, where we found an inverse correlation between the strength of intermolecular interactions that mediate the condensate formation and $D_{app}$, calculated from recovery plots obtained by FRAP analysis using 0.5% FITC-labeled peptides (Fig. 2b-c). WGR-3 has a slightly larger *D* than that of WGR-1, presumably because R-Y interactions are more favorable than R-F. WGR-4 has a lower diffusion coefficient, which suggests that the ELP domain induces partial intramolecular interactions which interfere sterically with the intermolecular interactions between the aromatic amino acid side chains, or between the aromatics and R. Decreasing the net charge of the peptide from +3 to +2 (WGR-5) significantly lowers the diffusion coefficient due to the decrease in electrostatic repulsion of the peptide molecules and increase in intermolecular interactions (Fig. 2b-c). Next, we studied how peptide composition affects the EE by using GFP, rhodamine B, and fluorescein as model fluorescent payloads (Fig. 2d). The EE of these payloads ranges between 72-98%, where the hydrophobic F-containing peptide WGR-3 has the lowest EE of GFP, and the most polar peptide, WGR-4, has the highest EE of GFP and rhodamine B, suggesting that the peptide interacts with the dye either electrostatically, by π-π, or cation-π interactions. WGR-1 has the highest EE of fluorescein. In addition to π-π interactions, WGR-1 might form H-bonding through the Y hydroxyl groups or electrostatic interactions with the deprotonated carboxylic acid of the dye. **Thus, these results demonstrate that the encapsulation efficiency of the condensates can be modulated by the chemical composition of the peptide building blocks.**

**(ii)** We recently developed peptide-RNA condensates that are inspired by viral factories, intracellular microcompartments which host the viral replication and assembly machinery (**Advanced Materials, 2022**)[15]. We focused on the measles virus as a model system and identified amino acids that are enriched in the disordered region of the measles phosphoprotein, which is responsible for phase separation and formation of viral factories. The designed viral factories-inspired peptide (VFP-1) is a 14-mer polar peptide: LGKSGRLPGKSGRV. Following systematic Ala scanning and identification that the basic amino acids are critical for LLPS, we sought to regulate the hydrophobicity in the condensates, and in turn, to control their material properties. For this, we substituted one of the K in the peptide with a lysine which is side chain-protected by the photocleavable group 6-nitroveratryloxy-carbonyl (Nvoc; peptide denoted Nvoc-VFP-1) and analyzed the effect of UV irradiation on the material properties of the condensates. We showed that the dynamics of the condensates can be regulated by irradiation time (Fig. 4a-b). Moreover, our results show that the encapsulation efficiency (EE) of rhodamine B (Fig. 4c-d) among other payloads[15] can be regulated by light, where selective irradiation of individual condensates results in 40% payload release (Fig. 4e). These results thus far **demonstrate that functionality of condensates i.e., their ability to encapsulate and release payloads can be regulated by the chemical composition of the peptide building block.** Thus, these results set the ground for more in-depth studies of how the chemical composition of condensates, the microenvironment hydrophobicity within the condensate, and its materials properties affect the partitioning of reactants and their chemical transformation.

Risk analysis strategy: (i). *Design challenge*: Our peptide design approach is bioinspired and based on rational design: incorporation of biological LLPS-promoting motifs (LCD motifs in IDPs) to the peptide sequence and integration with amino acids that promote intermolecular interactions and/or specific chemical functionality, e.g., H-bonding-promoting polar side chains, hydrophobic interactions-promoting side chains, or π-interactions-promoting side chains. This rational design is based on our results, knowledge and expertise in the design of self-assembling[51,52,54,55,62,63,75,76] or phase separating peptides[14,15,6356]. The challenge in this aim is to create condensates with controlled properties and functionality i.e., regulation of catalysis (which will be tested in Aim 2-3). To reduce the risk that the suggested libraries will not result in sufficient level of control, we will use coarse-grained simulations and atomistic simulations to provide molecular level insights on peptide-peptide/nucleic acid interactions that mediate condensate formation which can facilitate prediction of new peptide building blocks. The findings and insights obtained from this integrated characterization will directly feed peptide design optimization steps to control the properties of the condensates. (ii) *Characterization challenge*: Analyzing dynamic liquid assemblies is currently one of the major challenges in the field of bioinspired supramolecular chemistry[63]. To gain insights into the formation of peptide condensates by LLPS, we intend to rely on our extensive experience in characterization of peptide assemblies and to utilize various complementary microscopy and spectroscopy techniques as described in sections (b-e). Since some of these characterization tools (TEM, WAXS) provide information only on the dry-state, which might not accurately represent the condensates in solution, we will complement these analyses with techniques that allow solution-state analysis including confocal microscopy, FTIR, CD, Raman spectroscopy and microscopy (all available at TAU), NMR, which will provide a comprehensive

understanding of the intermolecular interactions underlying condensate formation. The results from these complementary analyses will feed back into the peptide design step. As described in the 'Design challenge' above, coarse-grained, and atomistic simulations will be also conducted to provide molecular level insights on the organization of peptides into condensates.

Impact and significance: I expect that the libraries constructed in this project will serve our scientific community for a variety of biotechnological applications including delivery of biomolecules, microreactors or sensors. I expect that the findings from the extensive characterization in this aim will provide insights into peptide, peptide-RNA, or peptide-biopolymer LLPS and thus advance our understanding of biological LLPS systems. Importantly, the intermediate goals of this aim will greatly promote our currently limited ability to regulate the properties of condensates with specific emphasis on condensates that serve as microreactors or reaction vessels.

**Aim 2**: Regulation of organic reactions by controlled microenvironments using designer condensates. **My underlying hypothesis** is that the capacity to spatially regulate the partitioning of hydrophobic organic molecules and their chemical transformation i.e., reaction rate and conversion, is determined by the chemical composition of the peptide building block, which in turn controls the condensate architecture and materials properties. We will **elucidate how modulating the chemical composition i.e., aromatic content (Library 1), hydrophobicity (Library 2), polarity (Library 3), the architecture (Library 4) and the material properties of condensates affect the partitioning or reactants, their conversion, and the reaction rate.** This hypothesis builds on the promise of the micellar catalysis approach, which allows organic reactions to occur in organic solvent-free aqueous media[77,78], but takes it to the next level of reaction regulation, as the designed condensates are much more versatile in their chemical composition and properties than the polymers used in micellar catalysis. Moreover, based on previous studies which demonstrate the acceleration of biocatalytic reactions in condensates[43,44,58,67,79–82], we consider that a major advantage of using the proposed technology over current biotechnological methods is that in addition to the organic solvent-free conditions this technology offers, the partitioning of the reactants in the condensates is expected to accelerate reaction kinetics, and thereby increase the reaction efficiency and reduce the manufacturing costs of the reagents. Due to the immense contribution of click chemistry to drug discovery and proteomic studies[83,84], we will focus on Cu(I)-catalyzed reactions between azide and terminal alkyne to form a 1,2,3-triazole, known as click reactions (reactions are detailed in Fig. 5). We will focus on **3 different click reactions as model systems**: (i) Reaction 1: synthesis of the fluorescent product 7-hydroxy-3-(4-(4-methoxyphenyl)-1H-1,2,3-triazol-1-yl)-2H-chromen-2-one; (ii) Reaction 2: synthesis of the Src kinase inhibitor[85] (4-phenyl-1H-1,2,3-triazol-1-yl)(thiophen-2-yl)methanone; and (iii) Reaction 3: synthesis of the anticancer xanthone derivative[86] 3-((1-(4-bromobenzyl)-1H-1,2,3-triazol-4-yl)methoxy)-1-hydroxy-9H-xanthen-9-one. These reactions were selected to allow a combination of a trackable fluorescent product (Reaction 1), and pharmaceutically active products, anticancer drugs (Reactions 2 and 3), which are biotechnologically relevant. Moreover, we selected reactants that have varying hydrophobicity (LogP=2.5 for Reactions 1 and 2; LogP=3.6 for Reaction 3), in order to tailor the condensates for reactants with varying level of hydrophobicity/polarity. The fluorescent product in Reaction 1 will allows us to spatially analyze how the core-shell architecture of condensates affect the reaction conversion and product localization in condensates (more details in Aim 2.4). All reactions are Cu(I)-catalyzed and occur in aqueous solution containing the designed peptide condensates as the reaction media. The suggested system is presented in Fig. 5.

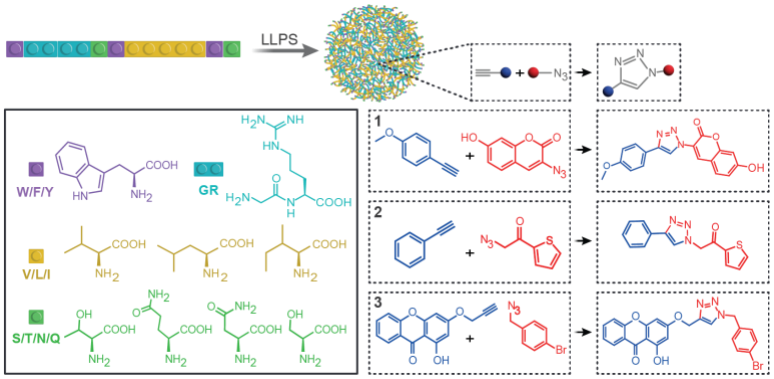


**Fig. 5. Designed condensates for regulation of organic reactions.** *Top panel:* libraries of peptide condensates will be designed by incorporating various motifs to the peptide sequences (detailed in Objective 1). Lego building blocks represent motifs of LLPS-promoting peptides. Lego color code (left panel): LLPS-promoting motifs GR (turquoise), aromatic (purple), aliphatic (yellow) and polar (green) amino acids. The condensates will host various click reactions as model systems. General illustration of the reaction is described in blue-red spheres. *Bottom right panel:* chemical structures of reactants and products of click model reactions.

2.1. Characterization and optimization of LLPS under reaction conditions: In the suggested reactions, sodium ascorbate (NaAsc) reduces Cu(II) to the active Cu(I) form which reacts with the ethynyl group, creating a complex that acts as a strong electrophile, enabling the azide group to attack the triple bond, and forming the triazole ring. The reaction requires moderately acidic pH, to allow the ethynyl group to connect to the Cu(I) ion and later, by

incorporating a proton, the triazole ring disconnects from the Cu(I) ion. In addition, the reaction should be performed under argon environment, to avoid the oxidation of Cu(I) to Cu(II) by the air. For each of the model reactions (Fig. 5), we will first find the optimal set of reaction conditions in bulk (without condensates), including concentration of each reactant, pH, NaAsc and Cu concentrations. We will utilize the libraries of condensates developed in Aim 1 (Table 1) to assess the overarching hypothesis. Following the synthesis and purification of peptides from Libraries 1-4, we will characterize the LLPS propensity and condensate formation under optimal conditions for each reaction. We will generate phase diagrams for each peptide as a function of pH ranging from 3-6 using turbidity assays and optical microscopy. We will also conduct phase diagrams as a function of NaAsc and Cu salts e.g., $CuSO_4$. The Cu salts serve both as catalysts and facilitate charge screening to initiate peptide LLPS. Next, we will label the peptides with a fluorescent dye (i.e., FITC) through the N-terminus amine group and use confocal laser scanning microscopy imaging to perform real-time imaging and FRAP studies to analyze the diffusivity of the peptides in the condensates.

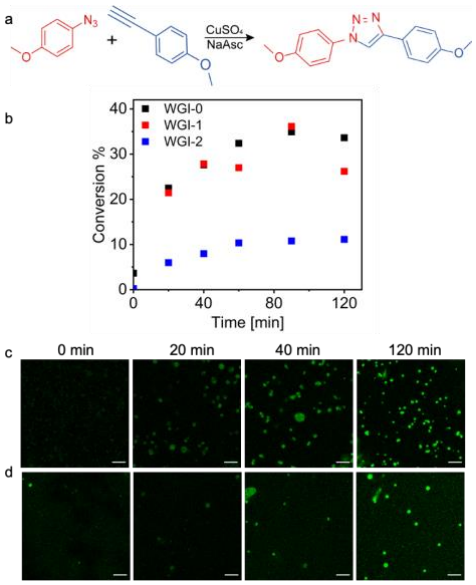


**Fig. 6. Regulation of organic reactions in peptide condensates (unpublished preliminary results:). a**. Click reaction used as a model system and formation of a fluorescent product. **b.** Kinetics of reaction conversion in WGI-0, WGI-1, and WGI-2 (10 mM) condensates. Reaction conversion was analyzed by HPLC over time. **c-d**. Spatial analysis of product formation in WGI-0 (**c**) or WGI-1 (**d**) condensates over time using confocal microscopy ($\lambda_{ex}$=405 nm, $\lambda_{em}$=465 nm). Scale bars=10 µm.

2.2. Partitioning analysis of reactants in condensates: To quantitatively analyze the partitioning of reactants in the designed condensates, we will use HPLC analysis to measure the concentration of each reactant in condensates based on calibration curves obtained in HPLC for each reactant. We will utilize the propensity of condensates to sediment following centrifugation and analyze the concentration of the reactants in the dilute phase (supernatant) following centrifugation. Encapsulated reactants are expected to be found in the pellet (condensed phase) which contain the condensates. This is a common method to quantify encapsulation efficiency (EE) and partitioning of payloads in condensates[14,15,53]. Furthermore, we will use biphasic coarse-grained molecular dynamics (MD) simulations (with the Dong lab) to analyze the partitioning of the reactants between the peptide condensate phase and the dilute phase. This method has proved to be very accurate for predicting LogP.[87] These analyses will help us to optimize peptide design in an iterative process (Fig. 1).

2.3. Quantitative analysis of reaction kinetics and conversion in condensates**:** We will analyze each of the click reactions in the designed peptide condensates (Libraries 1-4; Table 1) by systematically monitoring product formation in each of the designed condensates using HPLC/LC-MS. To obtain quantitative parameters i.e., % conversion over time (and thus obtain reaction rate) as a function of time using HPLC/LC-MS analysis, we will first purify the products from each reaction using preparative HPLC and obtain calibration curves of the products. To analyze the product, which is accumulated inside the condensates, all reactions will be quenched at specific time points by using 5% DMSO,25% DMF and 20% ACN before injected to the HPLC. The reactions in condensates will be compared with reactions in bulk containing organic solvents or reactions in bulk in aqueous media. Formation of the fluorescent product of Reaction 1 (Fig. 5) can be also monitored by kinetic analysis using absorbance/fluorescence spectroscopy, as complementary methods. The latter method relies on the spectral changes between the reactants and the click product of in Reaction 1 (and in a proof-of-concept click reaction described below in the preliminary results section).

2.4. Spatial analysis of reaction kinetics and product localization in condensates**:** The formation of fluorescent product in Reaction 1 and the spectral differences between the reactants and the click product enable spatial analysis of the reaction in condensates. We will analyze the formation of the product from Reaction 1 in condensates in real-time by laser scanning confocal microscopy to confirm that the product is indeed formed in the condensates. We will also analyze the condensates abundancy, morphology, and size over reaction time to examine how the reaction affects condensate stability and integrity. We will obtain reaction kinetics by monitoring the intensity of the fluorescent signal of the product in individual condensates over reaction time and compare the confocal microscopy kinetics analysis with that of the HPLC quantitative analysis. This analysis will provide important insights on the compartmentalized reaction inside the condensates vs. reaction in the dilute phase.

Notably, we will attempt to **regulate product localization in condensates with core-shell architecture** (Library 4). Specifically, we will analyze how the diameter of the core vs. shell affect the localization of the fluorescent product. We hypothesize that the product, which is more hydrophobic than the reactants (product LogP=2.78 vs. reactant LogP=1.76 and 2.49) will partition and be localized in the peptide-rich shell of the peptide-RNA condensates or in the peptide-rich core of the peptide-heparin condensates (Fig. 3) due to the higher hydrophobicity of the peptide-rich phases compared to that of the poly-U- or heparin-rich phases.

Preliminary results: We utilized the WGR-1 peptide as a template to form condensates with varying hydrophobicity. We studied 3 building blocks with varying number of hydrophobic amino acids: WGR-1 (denoted WGI-0), WGI-1 that has arginine/isoleucine (R3/I) substitution, and WGI-2, which has a R3/I and G6/I substitution. The condensates that are formed by simple coacervation of each peptide were used as reaction vessel to compartmentalize and regulate a model click reaction using 4-azidoanisole and 4-ethynylanisole as reactants, which results in the formation of the fluorescent product 1,4-bis(4-methoxyphenyl)-1H-1,2,3-triazole[88] (Fig. 6a). The product is expected to fluoresce owing to a three-ring conjugated $\pi$ system containing a resonance between all the three rings, and two electron donating groups, which enhance the resonance between the rings. We first confirmed condensate formation in the presence of the reactants using 5 mM of 4-ethynylanisole and 10 mM of 4-azidoanisole, where the latter is used at 2-fold molar excess to promote nucleophilic attack and product formation. Following optimization of reaction conditions, we confirmed that the peptide undergoes LLPS in citrate buffer at pH 5.5 and assemble into condensates in the presence of both reactants. We expected that Cu(I) catalyzes the reaction, while the $SO_4$ ions promote LLPS by charge screening of the basic peptide molecules. We monitored the conversion of the reaction in each condensate over time by using HPLC analysis (and using a calibration curve of the product). Interestingly, we found an inverse correlation between condensate hydrophobicity and reaction conversion, where increasing the number of isoleucine (I) in the peptide from 0 to 1 and from 0 to 2 decreased the conversion by 1.5-fold and 3-fold, respectively (Fig. 6b). The spectral difference between the reactants, which are not fluorescent at $\lambda_{ex}$=405 nm, and the click product, which is fluorescent ($\lambda_{ex}$=405 nm, $\lambda_{ex}$=465 nm), allowed us to spatially analyze reaction kinetics in condensates using confocal microscopy. This analysis confirmed that the reaction occurs in the condensates and that the kinetics in WGI-0 condensates (Fig. 6c) is faster than the kinetics in WGI-1 condensates (Fig. 6d).

Risk analysis strategy: (i) *LLPS under reaction conditions:* Since the reactants are hydrophobic and aromatic molecules, they might interfere with the peptide-peptide contacts that mediate LLPS and condensate formation, e.g., by $\pi$-interactions with the aromatic side chains of the peptide or might promote peptide aggregation and sedimentation. Therefore, we propose to first carefully examine the optimal reaction conditions and thereafter characterize peptide LLPS under these conditions (Aim 2.1). These characterization steps are crucial for the success of Aim 2. (ii) *Limited reactant partitioning and reaction yield:* For the condensates to act as reaction vessels, the reactants must partition in the condensates. However, due to the relatively high hydrophobicity of the reactants, there is a risk that they will have limited partition in the condensates and, as a result, the reaction conversion might drop. Our preliminary results show that the hydrophobicity of the peptide building block regulates reaction kinetics and conversion (Fig. 6b-c). To mitigate this risk, we propose to analyze the condensates in Libraries 1-4 which have varying number of aromatic, aliphatic and polar amino acids. By using this systematic approach, we expect to elucidate which motifs in the chemical composition increase reaction yield and reactant partitioning.

Impact and significance: The results from the intermediate goals of Aim 2 will provide insights into the structure-function relations of condensates that serve as reaction vessels. Specifically, we expect to shed light on how the condensate chemical and physical properties affect reactant partitioning and reaction conversion. In addition, this aim will result in the development of condensates that regulates the synthesis of chemotherapeutic anticancer drugs (Fig. 5). This approach can be further applied to different click reactions which result in various additional drugs. The broader impact of Aim 2 is the development of green chemistry technologies that will facilitate the use of biological building blocks for drug synthesis, as the technologies developed here can be further expanded to regulate organic reactions in organic solvent-free microenvironments.

**Aim 3:** Develop peptide condensates with emergent catalytic capacity and controlled reactivity. Supramolecular structures formed by simple self-assembling peptides have been shown to mimic the catalytic activity of hydrolases[75,89–92] through incorporation of histidine (H), either alone or bound to a metal ion cofactor, or in combination with additional amino acids such as serine and aspartic/glutamic acid that act as a catalytic triad[89,90]. These designs typically involve the incorporation of a single functional group, e.g., the imidazole of histidine, with self-assembly-prone amino acids, where the reactivity is improved by the intermolecular interactions between the

peptide building blocks. In contrast to catalytic self-assembling peptides, LLPS-promoting peptides are disordered and interact through weaker non-covalent interactions to form biomolecular condensates, which are inherently different from solid hierarchical self-assembled structures. Yet, synthetic biomolecular condensates are formed by multivalent non-covalent intermolecular interactions and can be designed to encapsulate molecules at >95% efficiency while keeping high water content, in comparison with solid hierarchical assemblies or aggregates. **My hypothesis is that condensates that are formed by peptide building blocks which contain catalytic moieties will have emergent catalytic capacity.** I envision that this emergent catalytic capacity is a result of the condensation and concentration of peptide building blocks with catalytic moieties in the condensate phase, concentration of substrates/reactants, and multiple modes of peptide-peptide and peptide-ion intermolecular interactions within the condensed phase**.**

We will focus on ester hydrolysis as a model system and develop condensates with capacity to enhance hydrolysis of ester by utilizing histidine (H) containing peptide building blocks (Library 5; Table 1). The suggested catalytic mechanism is a cofactor-dependent catalysis using $Zn^{2+}$, through coordination of the metal cofactor and the H within the peptide building blocks (Fig. 7, upper panel). In this context, we will systematically study how the number of H and their positions in the peptide sequence affect the catalytic activity. Based on our findings with peptide building blocks from Library 5 as the first step and informed by the findings from Aim 1, we will utilize peptide from Libraries 1-4 and systematically incorporate H to these sequences (the number and position of H will be revealed by the first round of analyses). We predict that our peptide design (Table 1, Library 5) will produce condensates that are mediated through multiple intermolecular interactions e.g., hydrogen bonding, hydrophobic interactions, and π-π stacking, similar to those of natural enzymes. Our preliminary results (Fig. 8) show that condensates that are formed by complex coacervation of cationic and anionic peptides have emergent catalytic capacity. Based on these promising results, we will employ both peptide simple coacervation and complex peptide-peptide or peptide/RNA coacervation for construction of condensates.

**The emergent catalytic activity of the condensates will be investigated by three ester hydrolysis model reactions (Fig. 7, lower panel):** (*i*) simple ester hydrolysis reaction of *p*-nitrophenyl acetate (*p*NPA), which generates the chromogenic product, *p*-nitrophenolate (*p*NP) that can be easily measured spectroscopically; (*ii*) ester hydrolysis of 8-acetoxypyrene-1,3,6-trisulfonate (pyranine acetate, PyOAc) to pyranine (PyOH), which turns fluorescent going from acidic to neutral pH[93] and can be both spectroscopically and microscopically analyzed, and (*iii*) hydrolysis of the prodrugs irinotecan to form the active oncolytic drug SN-38[94,95], commonly used for colon cancer therapy. These reactions were selected as they allow a spectroscopic readout and reaction monitoring (absorbance spectroscopy for *p*NP and fluorescence spectroscopy for PyOH), and due to their biotechnological and pharmaceutical relevancy as an anticancer drug (SN-38).

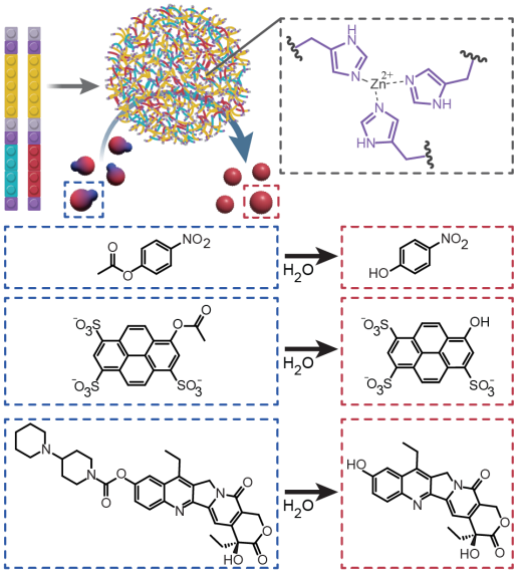


**Fig. 7. Proposed catalytic condensates system.** ***Upper panel:*** libraries of catalytic condensates that are formed by LLPS H-containing peptide will be developed. Lego building block color code: LLPS-promoting motif GR or GE (turquoise/red), aromatic amino acids (purple), histidine (grey). ***Inset from condensate*** showing the proposed catalytic mechanism involving three histidines from neighboring peptide molecules within the condensate bound to a single $Zn^{2+}$ ion. ***Lower panel:*** we will test our hypothesis using three ester hydrolysis model reactions (top to bottom): *p*NPA hydrolysis to *p*NP, PyOAc hydrolysis to PyOH, and irinotecan hydrolysis to SN-38.

3.1. Characterization and optimization of LLPS under reaction conditions: Based on the findings in Aim 1, we will select peptide building blocks from Library 5, which form liquid condensates as indicated by the FRAP analysis in Aim 1.4. Liquid condensates, as opposed to solid-like or gel-like condensates, have high water content that is critical for catalysis through the proposed mechanism (to create the nucleophile). We will start by analyzing 3 selected peptides (Library 5). Later on and based on the findings from Aim 3.1 and 3.2, we will expand peptide selection and incorporate H into peptides from Libraries 1-4. Following peptide synthesis and purification, we will generate phase diagrams for each peptide under the specific reaction condition for each of the three model reactions, and as a function of peptide concentration and Zn salt concentration.

3.2. Partitioning analysis of reactants in condensates: We will quantitatively analyze the partitioning of the substrates in reaction model system by absorbance measurements, fluorescence spectroscopy, and HPLC

analysis of samples following centrifugation, as detailed in Aim 2.2. Additionally, we will analyze the partitioning of the three substrates using biphasic coarse-grained MD simulations (with Dong lab).

3.3. Quantitative analysis of product formation and calculation of kinetic parameters: Catalytic activity and kinetic parameters including $K_{cat}$ and $K_m$ will be obtained by monitoring product formation over time using absorbance (*p*NP, at $\lambda$=405 nm) or fluorescence (PyOH, at $\lambda_{ex}$=454 nm and $\lambda_{em}$=517 nm) spectroscopy measurements, and HPLC analysis (for all three reactions). The latter will be critical for monitoring formation the non-chromogenic products SN-38. By varying the concentration of substrates and obtaining the initial velocity $V_0$ for each kinetic analysis and using the Michaelis Menten model, we will obtain kinetic parameters ($K_{cat}$, $K_m$) and compare these parameters and the catalytic efficiency of the condensates with reported parameters of peptide-based catalysts and natural enzymes. We expect that following optimization of condensate design, the catalytic efficiency of the condensates will exceed those of self-assembling peptides, which range between 0.17 to 3 $M^{-1}S^{-1}$[193,96,97], and that the $K_{cat}$ of the condensates to be close to those of esterases (up to 10 $sec^{-1}$).

3.4. Spatial analysis of product formation in condensates by confocal microscopy: The fluorescent PyOH product allows us to spatially analyze the catalytic activity of condensates by monitoring the formation of PyOH over time using confocal microscopy, as detailed in Aim 2.4, owing to the spectral difference between the substrate, PyOAc, which has a very low fluorescence at $\lambda_{ex}$=488 nm and $\lambda_{em}$=517 nm, and the product PyOH, which shows high emission intensity at these wavelengths. Condensate abundancy, morphology, and size over reaction time will be also analyzed. We will obtain reaction kinetics by monitoring the intensity of the fluorescence signal of the product in individual condensate over reaction time and compare the confocal microscopy kinetics analysis with that of the absorbance/fluoresce spectroscopy and HPLC quantitative analysis. This analysis will provide important insights on the compartmentalized reaction inside the condensates vs. reaction in the dilute phase.

3.5. Molecular level MD analysis to elucidate how condensate properties affect their catalytic performance: The Dong lab has previously show that the catalytic mechanism of H-containing self-assembling peptides involve activation of water molecule coordinated by metal ion bound to the active center to form a nucleophilic hydroxyl group[98,99]. In collaboration with the Dong lab, we will analyze the concentration of peptide and $Zn^{2+}$ inside the condensates, and the water content of condensates. We will subsequently analyze the effect of peptide sequence on the local configuration of the active center from the point of view of the electronic structure to understand its influence on the catalytic activity of the condensates.

Preliminary results: Our preliminary recent results (unpublished) show that condensates formed by complex coacervation of H-containing peptide WGRGRGRGWPGVGHGH and its anionic counterpart WGEGEGEGWPGVGHGH in the presence of $ZnCl_2$ enhance the hydrolysis of PyOAc to PyOH, with calculated $K_{cat}$ and $K_m$ of 0.00021 [$sec^{-1}$] and 9.6E-05 [M], respectively (Fig. 8b). Using confocal microscopy analysis, and by monitoring the fluorescence of the product PyOH at $\lambda_{ex}$=488 nm, $\lambda_{em}$=517 nm, we confirmed that the reaction occurs in the condensates (Fig. 8a). Interestingly, by using cationic and anionic peptides containing no H, 1H, or 2H, we show that the number of H on the condensates positively correlates with the condensate catalytic performance (Fig. 8c). These early attempts provide initial evidence for emergent catalytic capacity of condensates. We plan to fully develop catalytic condensate with tunable reactivity and gain mechanistic understanding of how the chemical and physical properties affect their catalytic performance.

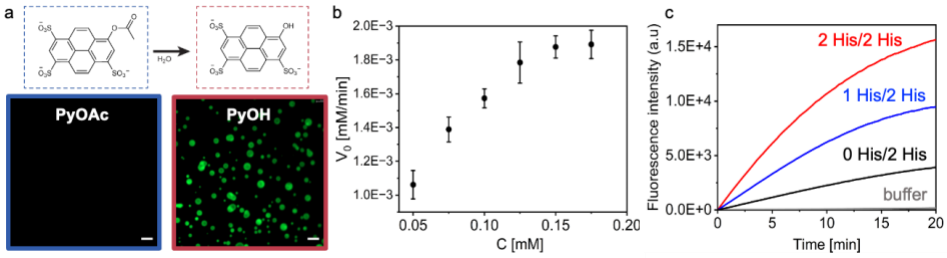


**Fig. 8. Peptide condensates with emergent catalytic capacity (Unpublished). a**. PyOAc ester hydrolysis to PyOH; confocal microscopy images of PyOH formation at t=0 min (left) and t=30 min (right) in catalytic condensates that are formed by the H2/H2 system. $\lambda_{ex}$=488 nm, $\lambda_{em}$=517 nm; Scale bars=5 µm. **b**. Michaelis Menten plot showing $V_0$ at increasing PyOAc concentrations for the H2/H2 condensates. **c**. Reaction kinetics of the H2/H2; H1/H2; H0/H2 condensates, PyOH formation was monitored by fluorescence spectroscopy ($\lambda_{ex}$=454 nm, $\lambda_{em}$=517 nm).

Risk analysis strategy: The ability of the condensates to catalyze ester hydrolysis of the three model substrates depends on substrate partitioning in the condensates, which might be limited due to the hydrophobicity and bulkiness of *p*NPA, irinotecan, and PyOAc, respectively. To address this, the number of aromatic and aliphatic/polar amino acids should be carefully considered, as incorporation of aromatic or aliphatic amino acids can promote substrate partitioning on one hand but might reduce metal ion concentration or water content in the

condensates, which in turn decreases catalytic efficiency. To mitigate this risk, we will harness the insights from Aim 1 on the exact composition of aromatic (Library 1) aliphatic/polar (Libraries 2-3) and H (Library 5) amino acids which results in LLPS and formation of dynamic condensates (rather than solid aggregation). We will implement these insights in Aim 3.1 and 3.2. In addition, based on previous work suggesting that R side chains in enzymes interact with sulfonate groups, we expect that the R found in the LLPS GR motif will promote interactions with PyOAc. Since the products of these ester hydrolysis reactions are more polar than the substrates, we expect them to have lower affinity to the peptides than the substrates, which should decrease the partitioning coefficient of the products in the condensates and promote their release. We will analyze this by time-dependent confocal microscopy analysis of the reactions in condensates.

Impact and significance: I expect to develop enzyme-free condensates that enhance catalysis and production of drugs. Since we plan to explore the mechanism of catalysis and to reveal how the peptide sequence affect catalytic performance, the findings from the intermediate goals of Aim 3 will provide important insights on the emergent properties of condensates and their function as catalysts. These insights will facilitate the development of condensates for nanotechnological applications, namely for green chemistry, and can be expanded for development of renewable energy technologies.

| **Year #** | **Year 1** | | **Year 2** | | **Year 3** | | **Year 4** | | **Year 5** | |
|---|---|---|---|---|---|---|---|---|---|---|
| **Month** | **Jan-Jun** | **Jul-Dec** | **Jan-Jun** | **Jul-Dec** | **Jan-Jun** | **Jul-Dec** | **Jan-Jun** | **Jul-Dec** | **Jan-Jun** | **Jul-Dec** |
| **Aim 1.1** | Peptide design and synthesis | | | | | | | | | |
| **Aim 1.2** | | Characterization of LLPS propensity | | | | | | | | |
| **Aim 1.3** | | Study condensate morphology and architecture | | | | | | | | |
| **Aim 1.4** | | Characterization of condensate material properties | | | | | | | | |
| **Aim 1.5** | | | Molecular-level characterization of condensates | | | | | | | |
| **Aim 2.1** | | | | LLPS under reac. cond. | | | | | | |
| **Aim 2.2** | | | | Reactant partitioning analysis | | | | | | |
| **Aim 2.3** | | | | Quantitative analysis of reaction kinetics and conversion | | | | | | |
| **Aim 2.4** | | | | | Spatial analysis of reaction kinetics in condensates | | | | | |
| **Aim 3.1** | | | LLPS under reaction conditions | | | | | | | |
| **Aim 3.2** | | | | Substrate partitioning analysis | | | | | | |
| **Aim 3.3** | | | | Quantitative kinetics analysis and catalytic performance of condensates | | | | | | |
| **Aim 3.4** | | | | | Spatial analysis of reaction kinetics in condensates | | | | | |
| **Aim 3.5** | | | | | | Elucidate catalytic mechanism of condensates by MS studies | | | | |

**Table 2**. **Suggested timeline of the project**. *CoRe* is designed as a five-year project. Aim 1 will start at the beginning of the 1st year and continue until the end of the 4th year. Based on insights from Aim 1, Aim 2 and Aim 3 will be pursued simultaneously and independently, starting from the 2nd year and until the end of the project on the end of the 5th year (see timeline in Table 2). Experimental findings, including molecular-level insights based on NMR (Aim 1-3), or material properties insights based on microscopy/spectroscopy analyses (Aim 1), or computational findings from simulations (Aim 1-3), will inform further design optimization of the building blocks (Aims 1-3), which will be iteratively analyzed (see Fig. 1 lower panel for the detailed workflow).

## Impact and vision - Future perspectives

The proposed project is high-risk/high-gain as if offers new approach to the far-reaching puzzle of reaction compartmentalization and mimicry of cellular organelles, as well as fundamentally new strategies for controlled reactivity in organic solvent-free environment for cell-free drug synthesis. **The combination of peptide design tools with a wide array of complementary characterization techniques will allow the fabrication of large libraries of condensates that are customized for specific reactions and tailored for drug synthesis.** I believe that my group has the right set of expertise and tools to succeed in this ambitious project. Our recently published results and on-going unpublished work demonstrate that advanced peptide design tools and characterization techniques can be leveraged for precisely control the properties of condensates for functionalities that go beyond the current state of the art. I am confident that we will make a significant impact on the field of LLPS, biomolecular self-assembly and biotechnology, and that the knowledge gained from this project will tremendously facilitate the development of condensates for emerging nanotechnologies.

**References**


(1) Banani, S. F.; Lee, H. O.; Hyman, A. A.; Rosen, M. K. Biomolecular Condensates: Organizers of Cellular Biochemistry. *Nat. Rev. Mol. Cell Biol.* **2017**, *18* (5), 285–298. DOI: 10.1038/nrm.2017.7.

(2) Dzuricky, M.; Rogers, B. A.; Shahid, A.; Cremer, P. S.; Chilkoti, A. De Novo Engineering of Intracellular Condensates Using Artificial Disordered Proteins. *Nat. Chem.* **2020**, *12* (9), 814–825. DOI: 10.1038/s41557-020-0511-7.

(3) Schuster, B. S.; Reed, E. H.; Parthasarathy, R.; Jahnke, C. N.; Caldwell, R. M.; Bermudez, J. G.; Ramage, H.; Good, M. C.; Hammer, D. A. Controllable Protein Phase Separation and Modular Recruitment to Form Responsive Membraneless Organelles. *Nat. Commun.* **2018**, *9* (1), 2985. DOI: 10.1038/s41467-018-05403-1.

(4) Shin, Y.; Berry, J.; Pannucci, N.; Haataja, M. P.; Toettcher, J. E.; Brangwynne, C. P. Spatiotemporal Control of Intracellular Phase Transitions Using Light-Activated OptoDroplets. *Cell* **2017**, *168* (1–2), 159-171.e14. DOI: 10.1016/j.cell.2016.11.054.

(5) Cummings, C. S.; Obermeyer, A. C. Phase Separation Behavior of Supercharged Proteins and Polyelectrolytes. *Biochemistry* **2018**, *57* (3), 314–323. DOI: 10.1021/acs.biochem.7b00990.

(6) Dai, Y.; You, L.; Chilkoti, A. Engineering Synthetic Biomolecular Condensates. *Nat. Rev. Bioeng.* **2023**, 1–15. DOI: 10.1038/s44222-023-00052-6.

(7) Roberts, S.; Dzuricky, M.; Chilkoti, A. Elastin-like Polypeptides as Models of Intrinsically Disordered Proteins. *FEBS Lett.* **2015**, *589* (19 Pt A), 2477–2486. DOI: 10.1016/j.febslet.2015.08.029.

(8) Blocher McTigue, W. C.; Perry, S. L. Design Rules for Encapsulating Proteins into Complex Coacervates. *Soft Matter* **2019**. DOI: 10.1039/C9SM00372J.

(9) Abbas, M.; Lipiński, W. P.; Wang, J.; Spruijt, E. Peptide-Based Coacervates as Biomimetic Protocells. *Chem. Soc. Rev.* **2021**, *50* (6), 3690–3705. DOI: 10.1039/d0cs00307g.

(10) Yuan, C.; Li, Q.; Xing, R.; Li, J.; Yan, X. Peptide Self-Assembly through Liquid-Liquid Phase Separation. *Chem* **2023**, *9* (9), 2425–2445. DOI: 10.1016/j.chempr.2023.05.009.

(11) Ma, L.; Fang, X.; Wang, C. Peptide-Based Coacervates in Therapeutic Applications. *Front. Bioeng. Biotechnol.* **2022**, *10*, 1100365. DOI: 10.3389/fbioe.2022.1100365.

(12) Aumiller, W. M.; Keating, C. D. Phosphorylation-Mediated RNA/Peptide Complex Coacervation as a Model for Intracellular Liquid Organelles. *Nat. Chem.* **2016**, *8* (2), 129–137. DOI: 10.1038/nchem.2414.

(13) Shen, Y.; Ruggeri, F. S.; Vigolo, D.; Kamada, A.; Qamar, S.; Levin, A.; Iserman, C.; Alberti, S.; George-Hyslop, P. S.; Knowles, T. P. J. Biomolecular Condensates Undergo a Generic Shear-Mediated Liquid-to-Solid Transition. *Nat. Nanotechnol.* **2020**, *15* (10), 841–847. DOI: 10.1038/s41565-020-0731-4.

(14) Baruch Leshem, A.; Sloan-Dennison, S.; Massarano, T.; Ben-David, S.; Graham, D.; Faulds, K.; Gottlieb, H. E.; Chill, J. H.; Lampel, A. Biomolecular Condensates Formed by Designer Minimalistic Peptides. *Nat. Commun.* **2023**, *14* (1), 421. DOI: 10.1038/s41467-023-36060-8.

(15) Katzir, I.; Haimov, E.; Lampel, A. Tuning the Dynamics of Viral-Factories-Inspired Compartments Formed by Peptide-RNA Liquid-Liquid Phase Separation. *Adv. Mater.* **2022**, *34* (47), e2206371. DOI: 10.1002/adma.202206371.

(16) Donau, C.; Späth, F.; Sosson, M.; Kriebisch, B. A. K.; Schnitter, F.; Tena-Solsona, M.; Kang, H.-S.; Salibi, E.; Sattler, M.; Mutschler, H.; Boekhoven, J. Active Coacervate Droplets as a Model for Membraneless Organelles and Protocells. *Nat. Commun.* **2020**, *11* (1), 5167. DOI: 10.1038/s41467-020-18815-9.

(17) Boeynaems, S.; Holehouse, A. S.; Weinhardt, V.; Kovacs, D.; Van Lindt, J.; Larabell, C.; Van Den Bosch, L.; Das, R.; Tompa, P. S.; Pappu, R. V.; Gitler, A. D. Spontaneous Driving Forces Give Rise to Protein-RNA Condensates with Coexisting Phases and Complex Material Properties. *Proc Natl Acad Sci USA* **2019**, *116* (16), 7889–7898. DOI: 10.1073/pnas.1821038116.

(18) Vieregg, J. R.; Lueckheide, M.; Marciel, A. B.; Leon, L.; Bologna, A. J.; Rivera, J. R.; Tirrell, M. V. Oligonucleotide-Peptide Complexes: Phase Control by Hybridization. *J. Am. Chem. Soc.* **2018**, *140* (5), 1632–1638. DOI: 10.1021/jacs.7b03567.

(19) Merindol, R.; Loescher, S.; Samanta, A.; Walther, A. Pathway-Controlled Formation of Mesostructured All-DNA Colloids and Superstructures. *Nat. Nanotechnol.* **2018**, *13* (8), 730–738. DOI: 10.1038/s41565-018-0168-1.

(20) Wang, X.; Shi, C.; Mo, J.; Xu, Y.; Wei, W.; Zhao, J. An Inorganic Biopolymer Polyphosphate Controls Positively Charged Protein Phase Transitions. *Angew. Chem. Int. Ed* **2020**, *59* (7), 2679–2683. DOI: 10.1002/anie.201913833.

(21) Bai, Q.; Liu, Z.; Chen, J.; Liang, D. Crowded Environment Regulates the Coacervation of Biopolymers via Nonspecific Interactions. *Biomacromolecules* **2023**, *24* (1), 283–293. DOI: 10.1021/acs.biomac.2c01129.

(22) Xue, S.; Zhou, F.; Zhao, T.; Zhao, H.; Wang, X.; Chen, L.; Li, J.-P.; Luo, S.-Z. Phase Separation on Cell Surface Facilitates BFGF Signal Transduction with Heparan Sulphate. *Nat. Commun.* **2022**, *13* (1), 1112. DOI: 10.1038/s41467-022-28765-z.

(23) Dammes, N.; Goldsmith, M.; Ramishetti, S.; Dearling, J. L. J.; Veiga, N.; Packard, A. B.; Peer, D. Conformation-Sensitive Targeting of Lipid Nanoparticles for RNA Therapeutics. *Nat. Nanotechnol.* **2021**, *16* (9), 1030–1038. DOI: 10.1038/s41565-021-00928-x.

(24) Krinsky, N.; Kaduri, M.; Zinger, A.; Shainsky-Roitman, J.; Goldfeder, M.; Benhar, I.; Hershkovitz, D.; Schroeder, A. Synthetic Cells Synthesize Therapeutic Proteins inside Tumors. *Adv. Healthc. Mater.* **2018**, *7* (9), e1701163. DOI: 10.1002/adhm.201701163.

(25) Large, D. E.; Abdelmessih, R. G.; Fink, E. A.; Auguste, D. T. Liposome Composition in Drug Delivery Design, Synthesis, Characterization, and Clinical Application. *Adv. Drug Deliv. Rev.* **2021**, *176*, 113851. DOI: 10.1016/j.addr.2021.113851.

(26) Lu, A.; O'Reilly, R. K. Advances in Nanoreactor Technology Using Polymeric Nanostructures. *Curr. Opin. Biotechnol.* **2013**, *24* (4), 639–645. DOI: 10.1016/j.copbio.2012.11.013.

(27) Tanner, P.; Baumann, P.; Enea, R.; Onaca, O.; Palivan, C.; Meier, W. Polymeric Vesicles: From Drug Carriers to Nanoreactors and Artificial Organelles. *Acc. Chem. Res.* **2011**, *44* (10), 1039–1049. DOI: 10.1021/ar200036k.

(28) Harman, C. L. G.; Patel, M. A.; Guldin, S.; Davies, G.-L. Recent Developments in Pickering Emulsions for Biomedical Applications. *Curr. Opin. Colloid Interface Sci* **2019**, *39*, 173–189. DOI: 10.1016/j.cocis.2019.01.017.

(29) Mane, S. R.; Sathyan, A.; Shunmugam, R. Biomedical Applications of PH-Responsive Amphiphilic Polymer Nanoassemblies. *ACS Appl. Nano Mater.* **2020**, *3* (3), 2104–2117. DOI: 10.1021/acsanm.0c00410.

(30) Vázquez-González, M.; Wang, C.; Willner, I. Biocatalytic Cascades Operating on Macromolecular Scaffolds and in Confined Environments. *Nat. Catal.* **2020**, *3* (3), 256–273. DOI: 10.1038/s41929-020-0433-1.

(31) Timmermans, S. B. P. E.; van Hest, J. C. M. Self-Assembled Nanoreactors Based on Peptides and Proteins. *Curr. Opin. Colloid Interface Sci* **2018**, *35*, 26–35. DOI: 10.1016/j.cocis.2018.01.005.

(32) Yewdall, N. A.; Mason, A. F.; van Hest, J. C. M. The Hallmarks of Living Systems: Towards Creating Artificial Cells. *Interface Focus* **2018**, *8* (5), 20180023. DOI: 10.1098/rsfs.2018.0023.

(33) Lim, J.; Kumar, A.; Low, K.; Verma, C. S.; Mu, Y.; Miserez, A.; Pervushin, K. Liquid-Liquid Phase Separation of Short Histidine- and Tyrosine-Rich Peptides: Sequence Specificity and Molecular Topology. *J. Phys. Chem. B* **2021**, *125* (25), 6776–6790. DOI: 10.1021/acs.jpcb.0c11476.

(34) Wei, W.; Petrone, L.; Tan, Y.; Cai, H.; Israelachvili, J. N.; Miserez, A.; Waite, J. H. An Underwater Surface-Drying Peptide Inspired by a Mussel Adhesive Protein. *Adv. Funct. Mater.* **2016**, *26* (20), 3496–3507. DOI: 10.1002/adfm.201600210.

(35) Pappu, R. V.; Cohen, S. R.; Dar, F.; Farag, M.; Kar, M. Phase Transitions of Associative Biomacromolecules. *Chem. Rev.* **2023**. DOI: 10.1021/acs.chemrev.2c00814.

(36) Martin, E. W.; Holehouse, A. S.; Peran, I.; Farag, M.; Incicco, J. J.; Bremer, A.; Grace, C. R.; Soranno, A.; Pappu, R. V.; Mittag, T. Valence and Patterning of Aromatic Residues Determine the Phase Behavior of Prion-like Domains. *Science* **2020**, *367* (6478), 694–699. DOI: 10.1126/science.aaw8653.

(37) Wang, J.; Choi, J.-M.; Holehouse, A. S.; Lee, H. O.; Zhang, X.; Jahnel, M.; Maharana, S.; Lemaitre, R.; Pozniakovsky, A.; Drechsel, D.; Poser, I.; Pappu, R. V.; Alberti, S.; Hyman, A. A. A Molecular Grammar Governing the Driving Forces for Phase Separation of Prion-like RNA Binding Proteins. *Cell* **2018**, *174* (3), 688-699.e16. DOI: 10.1016/j.cell.2018.06.006.

(38) Sun, Y.; Lau, S. Y.; Lim, Z. W.; Chang, S. C.; Ghadessy, F.; Partridge, A.; Miserez, A. Phase-Separating Peptides for Direct Cytosolic Delivery and Redox-Activated Release of Macromolecular Therapeutics. *Nat. Chem.* **2022**, *14* (3), 274–283. DOI: 10.1038/s41557-021-00854-4.

(39) Black, K. A.; Priftis, D.; Perry, S. L.; Yip, J.; Byun, W. Y.; Tirrell, M. Protein Encapsulation via Polypeptide Complex Coacervation. *ACS Macro Lett.* **2014**, *3* (10), 1088–1091. DOI: 10.1021/mz500529v.
(40) Dora Tang, T. Y.; van Swaay, D.; deMello, A.; Ross Anderson, J. L.; Mann, S. In Vitro Gene Expression within Membrane-Free Coacervate Protocells. *Chem. Commun.* **2015**, *51* (57), 11429–11432. DOI: 10.1039/c5cc04220h.
(41) Kato, S.; Garenne, D.; Noireaux, V.; Maeda, Y. T. Phase Separation and Protein Partitioning in Compartmentalized Cell-Free Expression Reactions. *Biomacromolecules* **2021**, *22* (8), 3451–3459. DOI: 10.1021/acs.biomac.1c00546.
(42) Schoenmakers, L. L. J.; Yewdall, N. A.; Lu, T.; André, A. A. M.; Nelissen, F. H. T.; Spruijt, E.; Huck, W. T. S. In Vitro Transcription-Translation in an Artificial Biomolecular Condensate. *ACS Synth. Biol.* **2023**, *12* (7), 2004–2014. DOI: 10.1021/acssynbio.3c00069.
(43) Küffner, A. M.; Prodan, M.; Zuccarini, R.; Capasso Palmiero, U.; Faltova, L.; Arosio, P. Acceleration of an Enzymatic Reaction in Liquid Phase Separated Compartments Based on Intrinsically Disordered Protein Domains. *ChemSystemsChem* **2020**. DOI: 10.1002/syst.202000001.
(44) Strulson, C. A.; Molden, R. C.; Keating, C. D.; Bevilacqua, P. C. RNA Catalysis through Compartmentalization. *Nat. Chem.* **2012**, *4* (11), 941–946. DOI: 10.1038/nchem.1466.
(45) Love, C.; Steinkühler, J.; Gonzales, D. T.; Yandrapalli, N.; Robinson, T.; Dimova, R.; Tang, T.-Y. D. Reversible PH-Responsive Coacervate Formation in Lipid Vesicles Activates Dormant Enzymatic Reactions. *Angew. Chem. Int. Ed* **2020**, *59* (15), 5950–5957. DOI: 10.1002/anie.201914893.
(46) Drobot, B.; Iglesias-Artola, J. M.; Le Vay, K.; Mayr, V.; Kar, M.; Kreysing, M.; Mutschler, H.; Tang, T.-Y. D. Compartmentalised RNA Catalysis in Membrane-Free Coacervate Protocells. *Nat. Commun.* **2018**, *9* (1), 3643. DOI: 10.1038/s41467-018-06072-w.
(47) Alberti, S.; Gladfelter, A.; Mittag, T. Considerations and Challenges in Studying Liquid-Liquid Phase Separation and Biomolecular Condensates. *Cell* **2019**, *176* (3), 419–434. DOI: 10.1016/j.cell.2018.12.035.
(48) Webber, M. J.; Berns, E. J.; Stupp, S. I. Supramolecular Nanofibers of Peptide Amphiphiles for Medicine. *Isr. J. Chem.* **2013**, *53* (8), 530–554. DOI: 10.1002/ijch.201300046.
(49) Lampel, A. Biology-Inspired Supramolecular Peptide Systems.
(50) Lampel, A.; Ulijn, R. V.; Tuttle, T. Guiding Principles for Peptide Nanotechnology through Directed Discovery. *Chem. Soc. Rev.* **2018**, *47* (10), 3737–3758. DOI: 10.1039/c8cs00177d.
(51) Lampel, A.; McPhee, S. A.; Park, H.-A.; Scott, G. G.; Humagain, S.; Hekstra, D. R.; Yoo, B.; Frederix, P. W. J. M.; Li, T.-D.; Abzalimov, R. R.; Greenbaum, S. G.; Tuttle, T.; Hu, C.; Bettinger, C. J.; Ulijn, R. V. Polymeric Peptide Pigments with Sequence-Encoded Properties. *Science* **2017**, *356* (6342), 1064–1068. DOI: 10.1126/science.aal5005.
(52) Lampel, A.; McPhee, S. A.; Kassem, S.; Sementa, D.; Massarano, T.; Aramini, J. M.; He, Y.; Ulijn, R. V. Melanin-Inspired Chromophoric Microparticles Composed of Polymeric Peptide Pigments. *Angew. Chem. Int. Ed* **2021**, *60* (14), 7564–7569. DOI: 10.1002/anie.202015170.
(53) Massarano, T.; Baruch Leshem, A.; Weitman, M.; Lampel, A. Spatiotemporal Control of Melanin Synthesis in Liquid Droplets. *ACS Appl. Mater. Interfaces* **2022**, *14* (18), 20520–20527. DOI: 10.1021/acsami.1c21006.
(54) Reddy, S. M. M.; Raßlenberg, E.; Sloan-Dennison, S.; Hesketh, T.; Silberbush, O.; Tuttle, T.; Smith, E.; Graham, D.; Faulds, K.; Ulijn, R. V.; Ashkenasy, N.; Lampel, A. Proton-Conductive Melanin-Like Fibers through Enzymatic Oxidation of a Self-Assembling Peptide. *Adv. Mater.* **2020**, *32* (46), e2003511. DOI: 10.1002/adma.202003511.
(55) Wulf, V.; Bichachi, E.; Hendler-Neumark, A.; Massarano, T.; Leshem, A. B.; Lampel, A.; Bisker, G. Multicomponent System of Single-Walled Carbon Nanotubes Functionalized with a Melanin-Inspired Material for Optical Detection and Scavenging of Metals. *Adv. Funct. Mater.* **2022**, 2209688. DOI: 10.1002/adfm.202209688.
(56) Netzer, A.; Katzir, I.; Baruch Leshem, A.; Weitman, M.; Lampel, A. Emergent Properties of Melanin-Inspired Peptide/RNA Condensates. *Proc Natl Acad Sci USA* **2023**, *120* (44), e2310569120. DOI: 10.1073/pnas.2310569120.
(57) Gaash, D.; Dewan, S.; Leshem, A. B.; Jaiswal, K. S.; Jelinek, R.; Lampel, A. Modulating the Optical Properties of Carbon Dots by Peptide Condensates. *Chem. Commun.* **2023**. DOI: 10.1039/d3cc03945e.

(58) Saha, B.; Chatterjee, A.; Reja, A.; Das, D. Condensates of Short Peptides and ATP for the Temporal Regulation of Cytochrome c Activity. *Chem. Commun.* **2019**, *55* (94), 14194–14197. DOI: 10.1039/c9cc07358b.
(59) Nakashima, K. K.; Vibhute, M. A.; Spruijt, E. Biomolecular Chemistry in Liquid Phase Separated Compartments. *Front. Mol. Biosci.* **2019**, *6*, 21. DOI: 10.3389/fmolb.2019.00021.
(60) Abbas, M.; Lipiński, W. P.; Nakashima, K. K.; Huck, W. T. S.; Spruijt, E. A Short Peptide Synthon for Liquid-Liquid Phase Separation. *Nat. Chem.* **2021**, *13* (11), 1046–1054. DOI: 10.1038/s41557-021-00788-x.
(61) Donau, C.; Späth, F.; Sosson, M.; Kriebisch, B.; Schnitter, F.; Tena-Solsona, M.; Kang, H.-S.; Salibi, E.; Sattler, M.; Mutschler, H.; Boekhoven, J. Active Coacervate Droplets as a Model for Membraneless Organelles and a Platform towards Synthetic Life. **2020**. DOI: 10.26434/chemrxiv.11648598.v1.
(62) Ulijn, R. V.; Lampel, A. Order/Disorder in Protein and Peptide-based Biomaterials. *Isr. J. Chem.* **2019**. DOI: 10.1002/ijch.201900051.
(63) Lampel, A. Biology-Inspired Supramolecular Peptide Systems. *Chem* **2020**, *6* (6), 1222–1236. DOI: 10.1016/j.chempr.2020.03.005.
(64) Nakashima, K. K.; van Haren, M. H. I.; André, A. A. M.; Robu, I.; Spruijt, E. Active Coacervate Droplets Are Protocells That Grow and Resist Ostwald Ripening. *Nat. Commun.* **2021**, *12* (1), 3819. DOI: 10.1038/s41467-021-24111-x.
(65) Späth, F.; Donau, C.; Bergmann, A. M.; Kränzlein, M.; Synatschke, C. V.; Rieger, B.; Boekhoven, J. Molecular Design of Chemically Fueled Peptide-Polyelectrolyte Coacervate-Based Assemblies. *J. Am. Chem. Soc.* **2021**, *143* (12), 4782–4789. DOI: 10.1021/jacs.1c01148.
(66) Fisher, R. S.; Elbaum-Garfinkle, S. Tunable Multiphase Dynamics of Arginine and Lysine Liquid Condensates. *Nat. Commun.* **2020**, *11* (1), 4628. DOI: 10.1038/s41467-020-18224-y.
(67) Liu, M.; He, S.; Cheng, L.; Qu, J.; Xia, J. Phase-Separated Multienzyme Biosynthesis. *Biomacromolecules* **2020**, *21* (6), 2391–2399. DOI: 10.1021/acs.biomac.0c00321.
(68) Wang, X.; Liu, X.; Huang, X. Bioinspired Protein-Based Assembling: Toward Advanced Life-Like Behaviors. *Adv. Mater.* **2020**, *32* (25), e2001436. DOI: 10.1002/adma.202001436.
(69) Capasso Palmiero, U.; Paganini, C.; Kopp, M. R. G.; Linsenmeier, M.; Küffner, A. M.; Arosio, P. Programmable Zwitterionic Droplets as Biomolecular Sorters and Model of Membraneless Organelles. *Adv. Mater.* **2021**, e2104837. DOI: 10.1002/adma.202104837.
(70) Gabryelczyk, B.; Cai, H.; Shi, X.; Sun, Y.; Swinkels, P. J. M.; Salentinig, S.; Pervushin, K.; Miserez, A. Hydrogen Bond Guidance and Aromatic Stacking Drive Liquid-Liquid Phase Separation of Intrinsically Disordered Histidine-Rich Peptides. *Nat. Commun.* **2019**, *10* (1), 5465. DOI: 10.1038/s41467-019-13469-8.
(71) Roberts, S.; Harmon, T. S.; Schaal, J. L.; Miao, V.; Li, K. J.; Hunt, A.; Wen, Y.; Oas, T. G.; Collier, J. H.; Pappu, R. V.; Chilkoti, A. Injectable Tissue Integrating Networks from Recombinant Polypeptides with Tunable Order. *Nat. Mater.* **2018**, *17* (12), 1154–1163. DOI: 10.1038/s41563-018-0182-6.
(72) Elsharkawy, S.; Al-Jawad, M.; Pantano, M. F.; Tejeda-Montes, E.; Mehta, K.; Jamal, H.; Agarwal, S.; Shuturminska, K.; Rice, A.; Tarakina, N. V.; Wilson, R. M.; Bushby, A. J.; Alonso, M.; Rodriguez-Cabello, J. C.; Barbieri, E.; Del Río Hernández, A.; Stevens, M. M.; Pugno, N. M.; Anderson, P.; Mata, A. Protein Disorder-Order Interplay to Guide the Growth of Hierarchical Mineralized Structures. *Nat. Commun.* **2018**, *9* (1), 2145. DOI: 10.1038/s41467-018-04319-0.
(73) Kato, M.; Han, T. W.; Xie, S.; Shi, K.; Du, X.; Wu, L. C.; Mirzaei, H.; Goldsmith, E. J.; Longgood, J.; Pei, J.; Grishin, N. V.; Frantz, D. E.; Schneider, J. W.; Chen, S.; Li, L.; Sawaya, M. R.; Eisenberg, D.; Tycko, R.; McKnight, S. L. Cell-Free Formation of RNA Granules: Low Complexity Sequence Domains Form Dynamic Fibers within Hydrogels. *Cell* **2012**, *149* (4), 753–767. DOI: 10.1016/j.cell.2012.04.017.
(74) Crowley, P. B.; Golovin, A. Cation-Pi Interactions in Protein-Protein Interfaces. *Proteins* **2005**, *59* (2), 231–239. DOI: 10.1002/prot.20417.
(75) Zhang, C.; Shafi, R.; Lampel, A.; MacPherson, D.; Pappas, C. G.; Narang, V.; Wang, T.; Maldarelli, C.; Ulijn, R. V. Switchable Hydrolase Based on Reversible Formation of Supramolecular Catalytic Site Using a Self-assembling Peptide. *Angew. Chem.* **2017**, *129* (46), 14703–14707. DOI: 10.1002/ange.201708036.
(76) Alakpa, E.; Jayawarna, V.; Lampel, A.; Burgess, K.; West, C.; Bakker, S. J.; Roy, S.; Javid, N.; Fleming, S.; Lamprou, D.; Yang, J.; Miller, A.; Urquhart, A.; Frederix, P. J. M.; Hunt, N.; Péault, B.; Ulijn, R.; Dalby,

M. Tunable Supramolecular Hydrogels for Selection of Lineage-Guiding Metabolites in Stem Cell Cultures. *Chem* **2016**, *1* (2), 298–319. DOI: 10.1016/j.chempr.2016.07.001.

(77) Shen, T.; Zhou, S.; Ruan, J.; Chen, X.; Liu, X.; Ge, X.; Qian, C. Recent Advances on Micellar Catalysis in Water. *Adv. Colloid Interface Sci.* **2021**, *287*, 102299. DOI: 10.1016/j.cis.2020.102299.

(78) Cortes-Clerget, M.; Yu, J.; Kincaid, J. R. A.; Walde, P.; Gallou, F.; Lipshutz, B. H. Water as the Reaction Medium in Organic Chemistry: From Our Worst Enemy to Our Best Friend. *Chem. Sci.* **2021**, *12* (12), 4237–4266. DOI: 10.1039/d0sc06000c.

(79) Chen, Y.; Yuan, M.; Zhang, Y.; Liu, S.; Yang, X.; Wang, K.; Liu, J. Construction of Coacervate-in-Coacervate Multi-Compartment Protocells for Spatial Organization of Enzymatic Reactions. *Chem. Sci.* **2020**, *11* (32), 8617–8625. DOI: 10.1039/d0sc03849k.

(80) Le Vay, K.; Song, E. Y.; Ghosh, B.; Tang, T.-Y. D.; Mutschler, H. Enhanced Ribozyme-Catalyzed Recombination and Oligonucleotide Assembly in Peptide-RNA Condensates. *Angew. Chem. Int. Ed* **2021**, *60* (50), 26096–26104. DOI: 10.1002/anie.202109267.

(81) Sokolova, E.; Spruijt, E.; Hansen, M. M. K.; Dubuc, E.; Groen, J.; Chokkalingam, V.; Piruska, A.; Heus, H. A.; Huck, W. T. S. Enhanced Transcription Rates in Membrane-Free Protocells Formed by Coacervation of Cell Lysate. *Proc Natl Acad Sci USA* **2013**, *110* (29), 11692–11697. DOI: 10.1073/pnas.1222321110.

(82) Liu, S.; Zhang, Y.; Li, M.; Xiong, L.; Zhang, Z.; Yang, X.; He, X.; Wang, K.; Liu, J.; Mann, S. Enzyme-Mediated Nitric Oxide Production in Vasoactive Erythrocyte Membrane-Enclosed Coacervate Protocells. *Nat. Chem.* **2020**, *12* (12), 1165–1173. DOI: 10.1038/s41557-020-00585-y.

(83) Thirumurugan, P.; Matosiuk, D.; Jozwiak, K. Click Chemistry for Drug Development and Diverse Chemical-Biology Applications. *Chem. Rev.* **2013**, *113* (7), 4905–4979. DOI: 10.1021/cr200409f.

(84) Parker, C. G.; Pratt, M. R. Click Chemistry in Proteomic Investigations. *Cell* **2020**, *180* (4), 605–632. DOI: 10.1016/j.cell.2020.01.025.

(85) Kumar, D.; Reddy, V. B.; Kumar, A.; Mandal, D.; Tiwari, R.; Parang, K. Click Chemistry Inspired One-Pot Synthesis of 1,4-Disubstituted 1,2,3-Triazoles and Their Src Kinase Inhibitory Activity. *Bioorg. Med. Chem. Lett.* **2011**, *21* (1), 449–452. DOI: 10.1016/j.bmcl.2010.10.121.

(86) Wu, J.; Dai, J.; Zhang, Y.; Wang, J.; Huang, L.; Ding, H.; Li, T.; Zhang, Y.; Mao, J.; Yu, S. Synthesis of Novel Xanthone Analogues and Their Growth Inhibitory Activity against Human Lung Cancer A549 Cells. *Drug Des. Devel. Ther.* **2019**, *13*, 4239–4246. DOI: 10.2147/DDDT.S217827.

(87) Marrink, S. J.; Risselada, H. J.; Yefimov, S.; Tieleman, D. P.; de Vries, A. H. The MARTINI Force Field: Coarse Grained Model for Biomolecular Simulations. *J. Phys. Chem. B* **2007**, *111* (27), 7812–7824. DOI: 10.1021/jp071097f.

(88) Meisner, Q. J.; Accardo, J. V.; Hu, G.; Clark, R. J.; Jiang, D.-E.; Zhu, L. Fluorescence of Hydroxyphenyl-Substituted “Click” Triazoles. *J. Phys. Chem. A* **2018**, *122* (11), 2956–2973. DOI: 10.1021/acs.jpca.8b00577.

(89) Duncan, K. L.; Ulijn, R. V. Short Peptides in Minimalistic Biocatalyst Design. *Biocatalysis* **2015**, *1* (1), 67–81. DOI: 10.1515/boca-2015-0005.

(90) Zozulia, O.; Dolan, M. A.; Korendovych, I. V. Catalytic Peptide Assemblies. *Chem. Soc. Rev.* **2018**, *47* (10), 3621–3639. DOI: 10.1039/c8cs00080h.

(91) Rufo, C. M.; Moroz, Y. S.; Moroz, O. V.; Stöhr, J.; Smith, T. A.; Hu, X.; DeGrado, W. F.; Korendovych, I. V. Short Peptides Self-Assemble to Produce Catalytic Amyloids. *Nat. Chem.* **2014**, *6* (4), 303–309. DOI: 10.1038/nchem.1894.

(92) Makam, P.; Yamijala, S. S. R. K. C.; Tao, K.; Shimon, L. J. W.; Eisenberg, D. S.; Sawaya, M. R.; Wong, B. M.; Gazit, E. Non-Proteinaceous Hydrolase Comprised of a Phenylalanine Metallo-Supramolecular Amyloid-like Structure. *Nat. Catal.* **2019**, *2* (11), 977–985. DOI: 10.1038/s41929-019-0348-x.

(93) Wang, P. S. P.; Nguyen, J. B.; Schepartz, A. Design and High-Resolution Structure of a $B^3$-Peptide Bundle Catalyst. *J. Am. Chem. Soc.* **2014**, *136* (19), 6810–6813. DOI: 10.1021/ja5013849.

(94) Laizure, S. C.; Herring, V.; Hu, Z.; Witbrodt, K.; Parker, R. B. The Role of Human Carboxylesterases in Drug Metabolism: Have We Overlooked Their Importance? *Pharmacotherapy* **2013**, *33* (2), 210–222. DOI: 10.1002/phar.1194.

(95) Sanghani, S. P.; Quinney, S. K.; Fredenburg, T. B.; Davis, W. I.; Murry, D. J.; Bosron, W. F. Hydrolysis of Irinotecan and Its Oxidative Metabolites, 7-Ethyl-10-[4-N-(5-Aminopentanoic Acid)-1-Piperidino] Carbonyloxycamptothecin and 7-Ethyl-10-[4-(1-Piperidino)-1-Amino]-Carbonyloxycamptothecin, by

Human Carboxylesterases CES1A1, CES2, and a Newly Expressed Carboxylesterase Isoenzyme, CES3. *Drug Metab. Dispos.* **2004**, *32* (5), 505–511. DOI: 10.1124/dmd.32.5.505.

(96) Broo, K. S.; Nilsson, H.; Nilsson, J.; Flodberg, A.; Baltzer, L. Cooperative Nucleophilic and General-Acid Catalysis by the $HisH^+$ −His Pair and Arginine Transition State Binding in Catalysis of Ester Hydrolysis Reactions by Designed Helix−Loop−Helix Motifs. *J. Am. Chem. Soc.* **1998**, *120* (17), 4063–4068. DOI: 10.1021/ja9737580.

(97) Javor, S.; Delort, E.; Darbre, T.; Reymond, J.-L. A Peptide Dendrimer Enzyme Model with a Single Catalytic Site at the Core. *J. Am. Chem. Soc.* **2007**, *129* (43), 13238–13246. DOI: 10.1021/ja074115f.

(98) Song, R.; Wu, X.; Xue, B.; Yang, Y.; Huang, W.; Zeng, G.; Wang, J.; Li, W.; Cao, Y.; Wang, W.; Lu, J.; Dong, H. Principles Governing Catalytic Activity of Self-Assembled Short Peptides. *J. Am. Chem. Soc.* **2019**, *141* (1), 223–231. DOI: 10.1021/jacs.8b08893.

(99) Chen, Y.; Yang, Y.; Orr, A. A.; Makam, P.; Redko, B.; Haimov, E.; Wang, Y.; Shimon, L. J. W.; Rencus-Lazar, S.; Ju, M.; Tamamis, P.; Dong, H.; Gazit, E. Self-Assembled Peptide Nano-Superstructure towards Enzyme Mimicking Hydrolysis. *Angew. Chem. Int. Ed* **2021**, *60* (31), 17164–17170. DOI: 10.1002/anie.202105830.

***Appendix: All current grants and on-going / submitted grant applications of the PI***
**CORE**

**Current research grants**

| Project title | Funding source | Amount (Euros) | Period | Role of the PI | Relation to current ERC proposal |
|---|---|---|---|---|---|
| *Expanding the matrix space – modulating growth factor signals by cell instructive GAG-peptide hydrogels with customizable viscoelastic properties* | DFG | 464,300 | 2021-2024 | Co-PI | None |
| *Spatiotemporal control melanin-inspired photoprotective materials* | Israel Science Foundation (ISF) | 352,902 | 2021-2025 | PI | None |
| *Systems chemistry: from biomolecules and molecular materials confernece* | Israel Science Foundation (ISF) grant for organization of conferences | 16,310 | 2023-2024 | Primary co-PI | None |
| *Systems chemistry: from biomolecules and molecular materials confernece* | TAU funding for conference organization | 4,660 | 2023-2024 | PI | None |
| *TWIN2PIPSA in collaboration of Uni. of Lisbon* | EU-funded project | None | 2023-2027 | Active staff member | None |

**Past research grants**

| Project title | Funding source | Amount (Euros) | Period | Role of the PI | Relation to current ERC proposal |
|---|---|---|---|---|---|
| *Spatiotemporal control melanin-inspired photoprotective materials* | Israel Science Foundation (ISF) – Equipment grant for new faculty | 8,428 | 2021-2022 | PI | None |
| *Melanin liquid droplets* | Israel Innovation Authority | 200,130 | 2021-2022 | PI | None |
| *Peptide compartments for cell-free biosynthesis of drugs* | Colton-Nadal Foundation | 40,456 | 2021-2022 | PI | None |

**On-going/submitted grant applications**

| Project title | Funding source | Amount (Euros) | Period | Role of the PI | Relation to current ERC proposal |
|---|---|---|---|---|---|
| *Elucidating the assembly properties and catalytic function of ordered/disordered peptide assemblies using integrated experimental and computational tools* | Israel Science Foundation (ISF)-NSFC join grant | 248,402 | 2024-2027 | Co-PI | Partial: Related to Aim 3, but significantly differ in the approach (design of self-assembling peptides). |
| *Chemical transformations in designer peptide condensates* | Volkswagen Foundation | 270,000 | 2024-2027 | Co-PI | Partial: Related to Aim 2 yet is focused on monitoring reactions by NMR. |